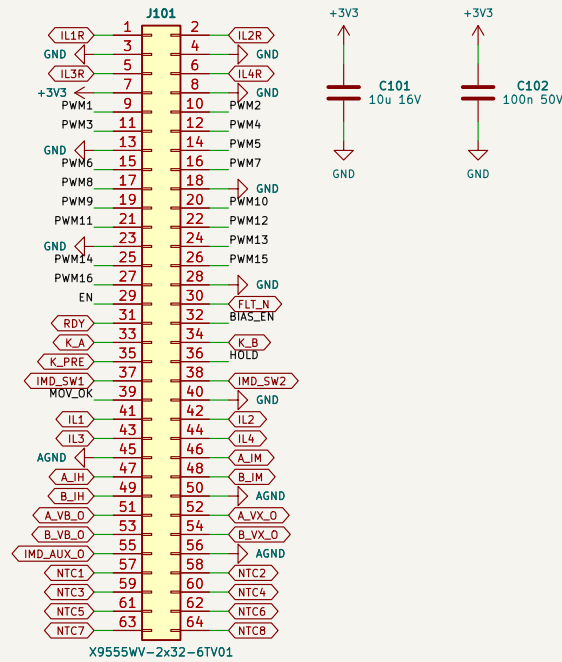
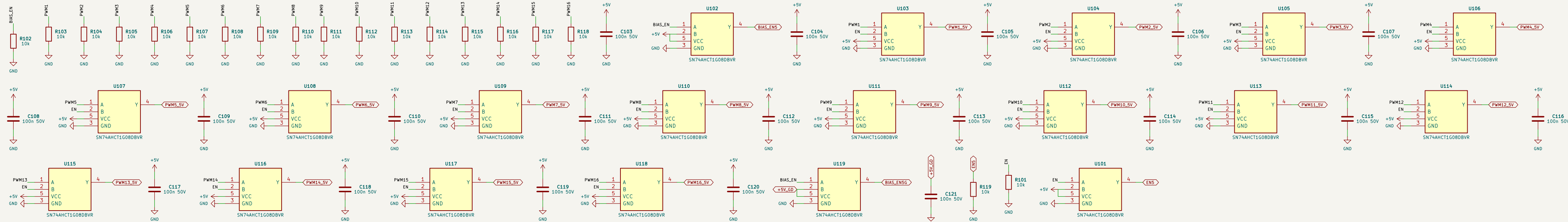


PC CONNECTOR (INTERFACES.PC), LIVE



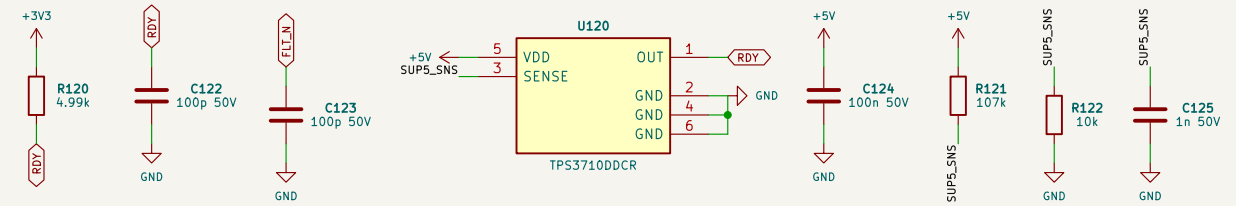
Both boards live: GND = AGND = BUS-; +3V3 goes TO the control board (rev A2: the former +24V / +5V pins carry the sensor references (ILR-IL4R)). Commands are 3.3 V logic after the control board's trip latch: every command input is pulled low here (open = off).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



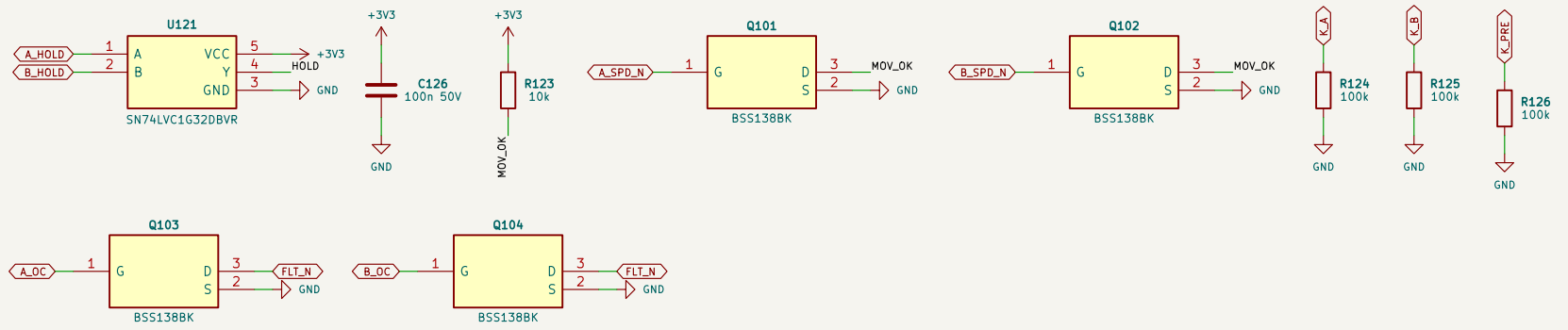
NSI6651 inputs are specified at VCC1 = 5 V only (VIN1 <= 3.3 V, >7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48–0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



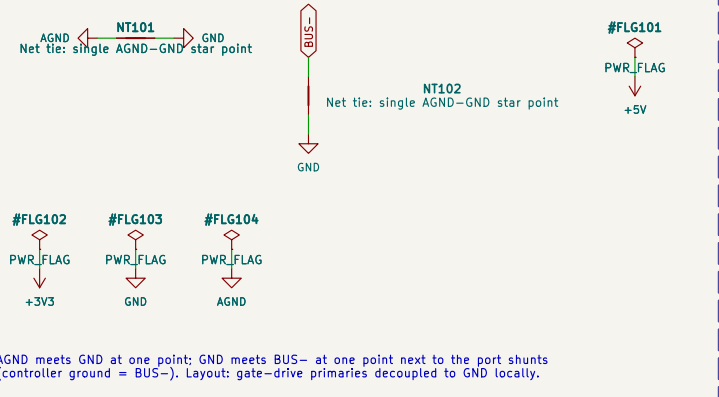
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC (I_{IL} >= 387–413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (R_{AF}=fuse gap, port_spec). K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N



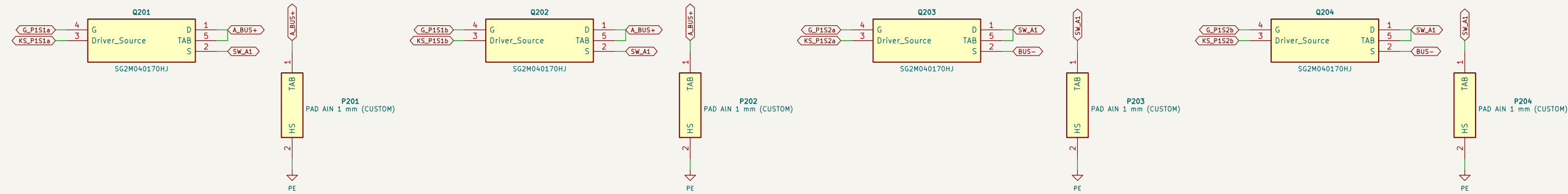
HOLD = A_HOLD OR B_HOLD (|I_{port}| >= 0.97–1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC (I_{IL} >= 387–413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (R_{AF}=fuse gap, port_spec). K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



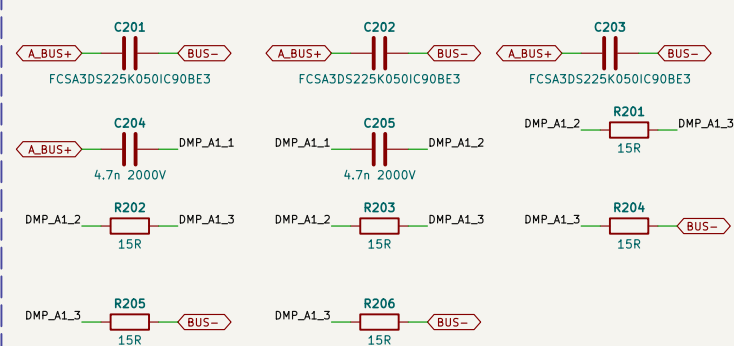
AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-). Layout gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



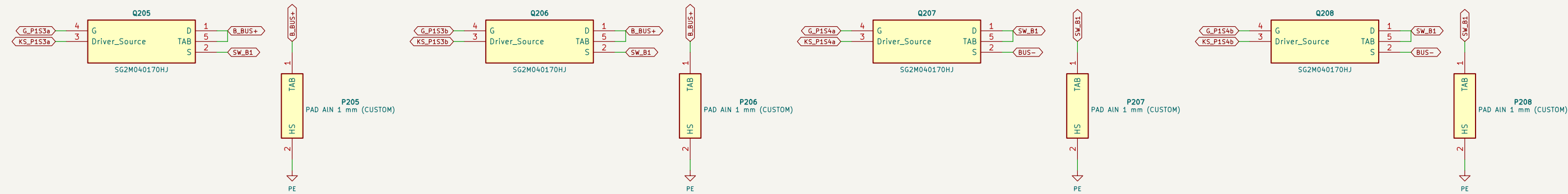
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AlN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



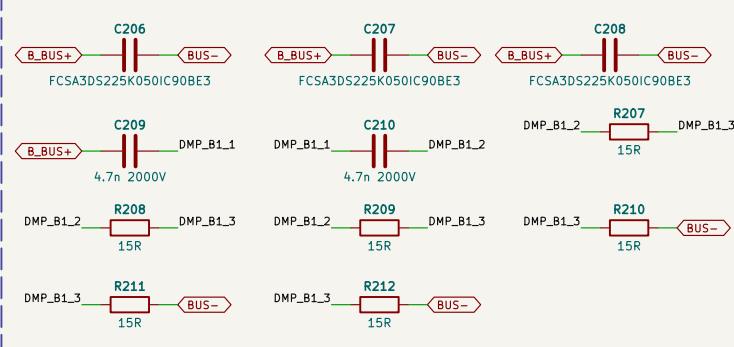
3 x FCSA3DS225 at the S1/S2 pins: commutation loop ≤ 20 nH, bank ≤ 15 nH from the leg (LAYOUT, leg deck)
 Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins ≤ 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



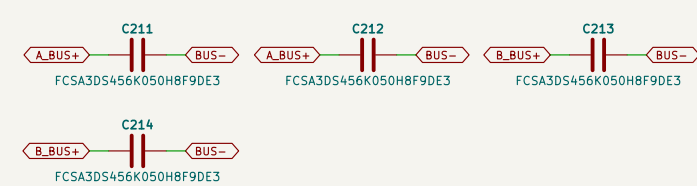
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



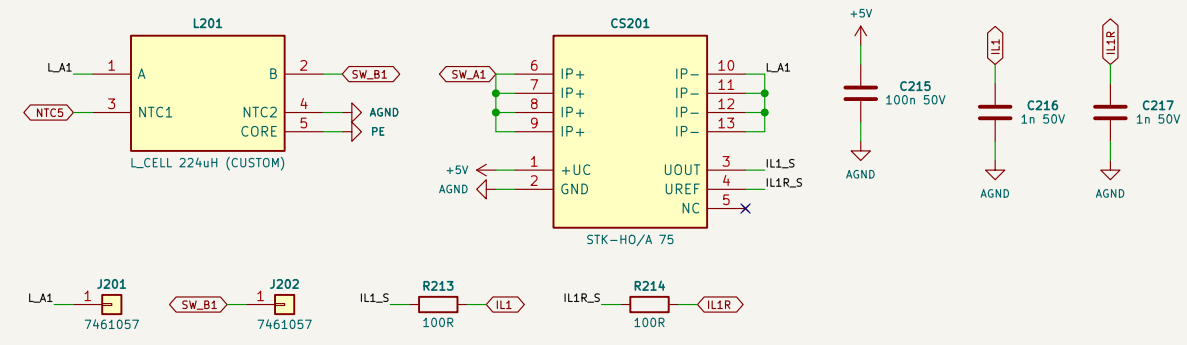
3 x FCSA3DS225 at the S3/S4 pins: commutation loop ≤ 20 nH, bank ≤ 15 nH from the leg (LAYOUT, leg deck)
 Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins ≤ 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



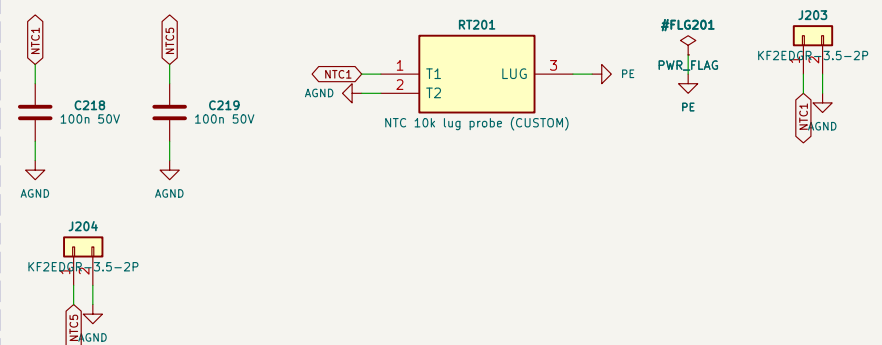
2 x FCSA3DS456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



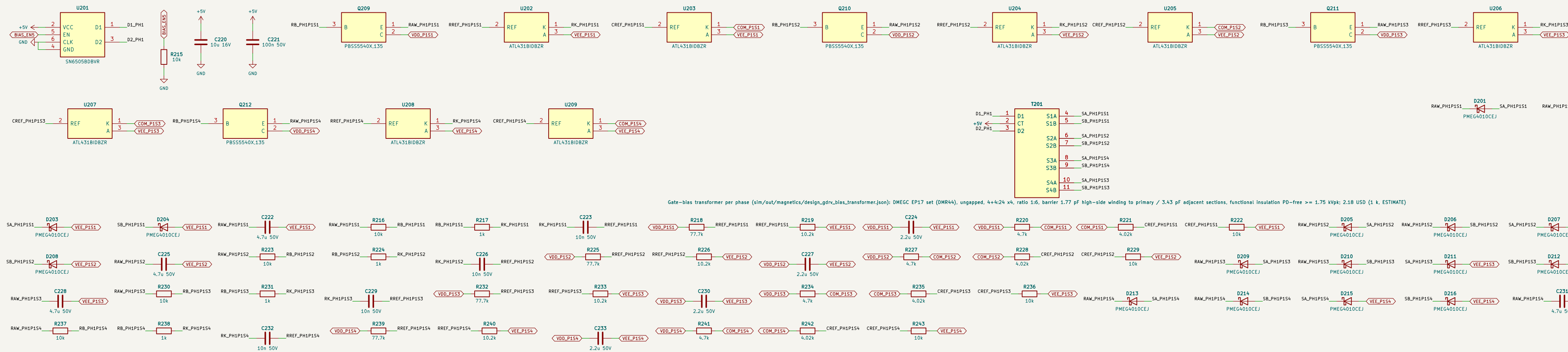
Lead A from SW_A1 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW_A -> SW_B) to stud L_A1; lead B on stud SW_B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A; 100R + 1 nF at the source. IL1R = the sensor's own Uref (pin 4). 100R + 1 nF: PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, ≥ 2.5 kV rms lead-to-lug (R-10)
NTC5 = inductor winding NTC (functional to the winding) 100 nF at the header against switch-node noise

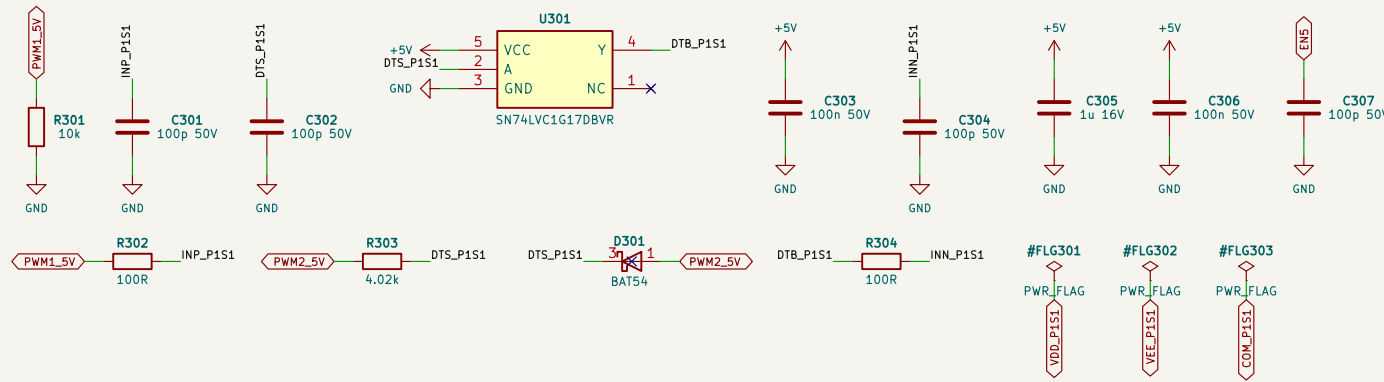
PH1: GATE BIAS (PER PHASE)



SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation (PD-free ≥ 1.75 kVp, ≤ 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split

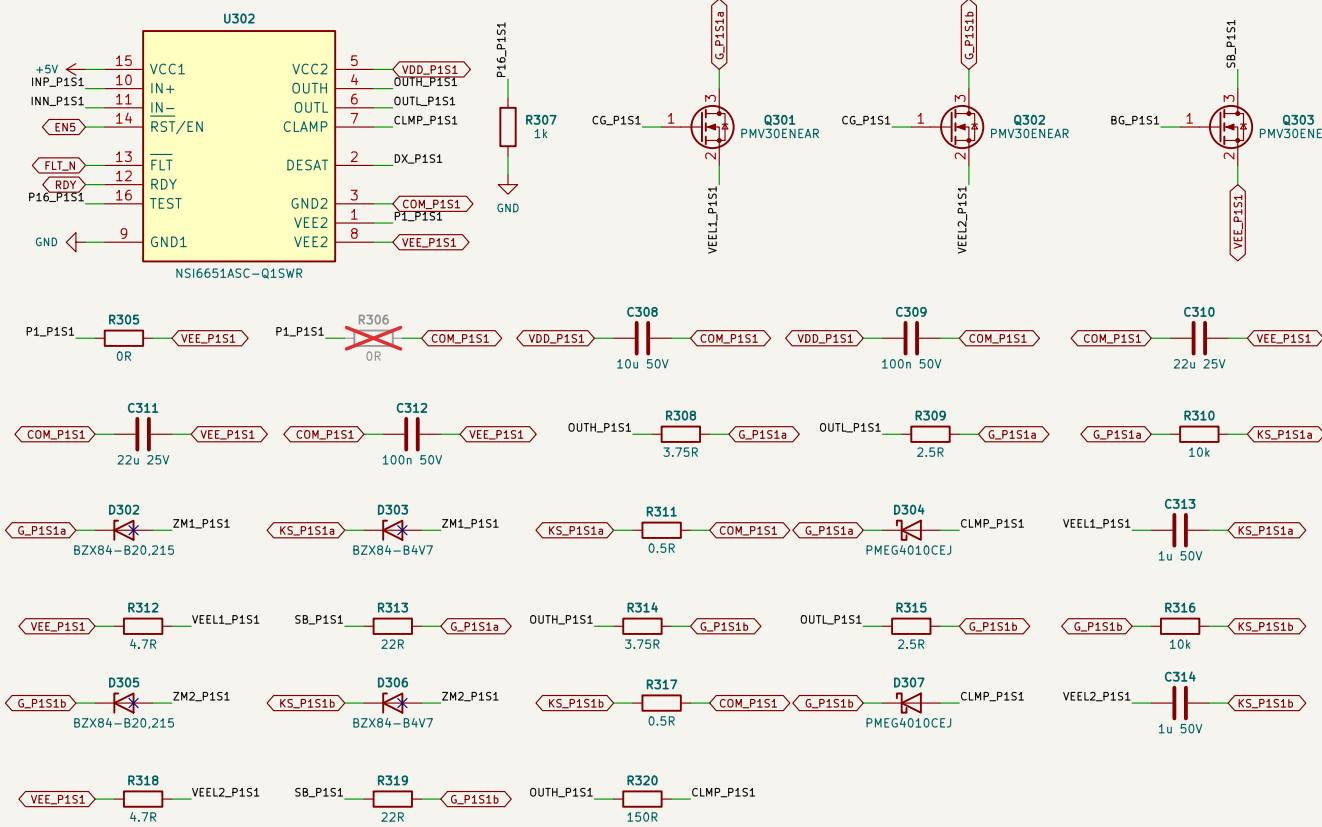
Not bench-validated. Values read from sim/out/*.json; checks: design_check()		
Sheet 3		
Schematics authored by Chinmoy Bhuyan		
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HW-PE bus		
Sheet: /02 Phase 1 power stage, bias, sensing/ File: 02_ph1_power.kicad_sch		
Title: PV-PWR-4 power board - Phase 1 power stage, bias, sensing		
Size: A2	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 3/26

P1S1: LOGIC SIDE



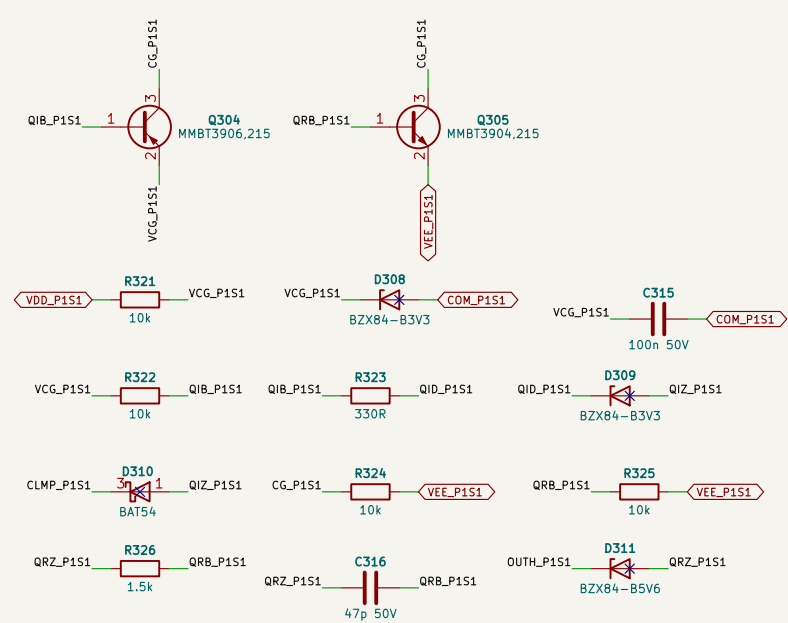
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S1: DRIVER AND 2 GATES



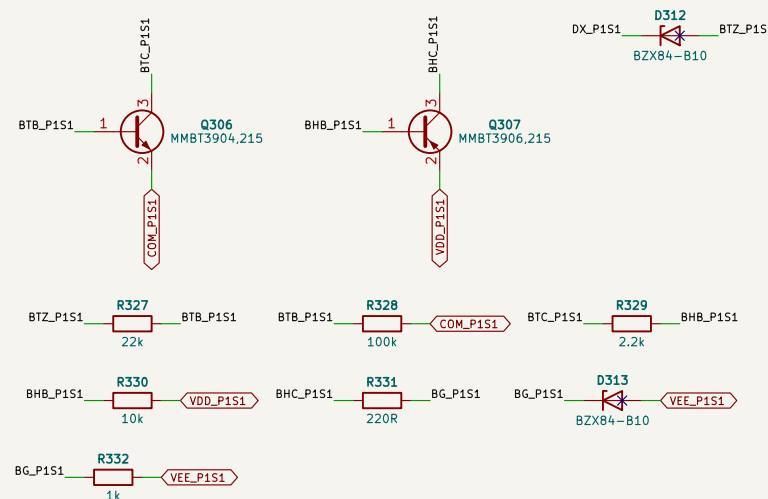
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S1: MILLER CLAMP CONTROL



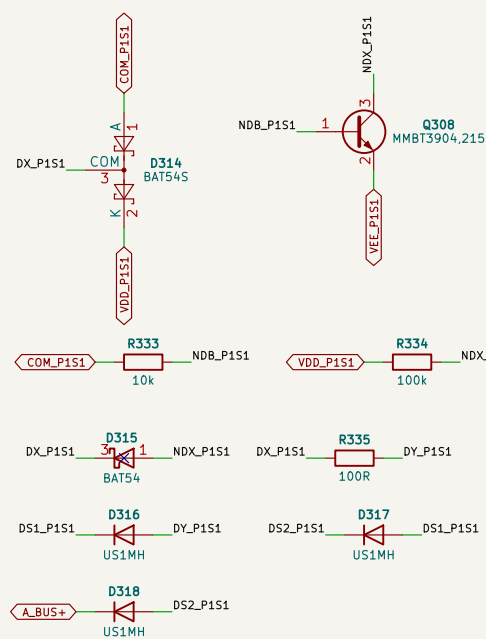
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S1: SHORT-CIRCUIT BOOSTER



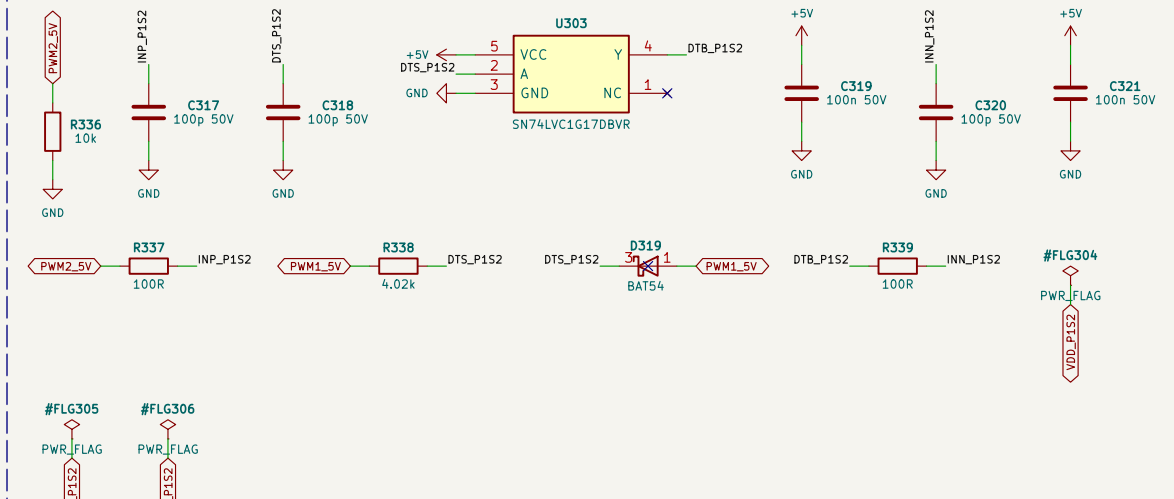
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ340DA + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P1S1: DESAT



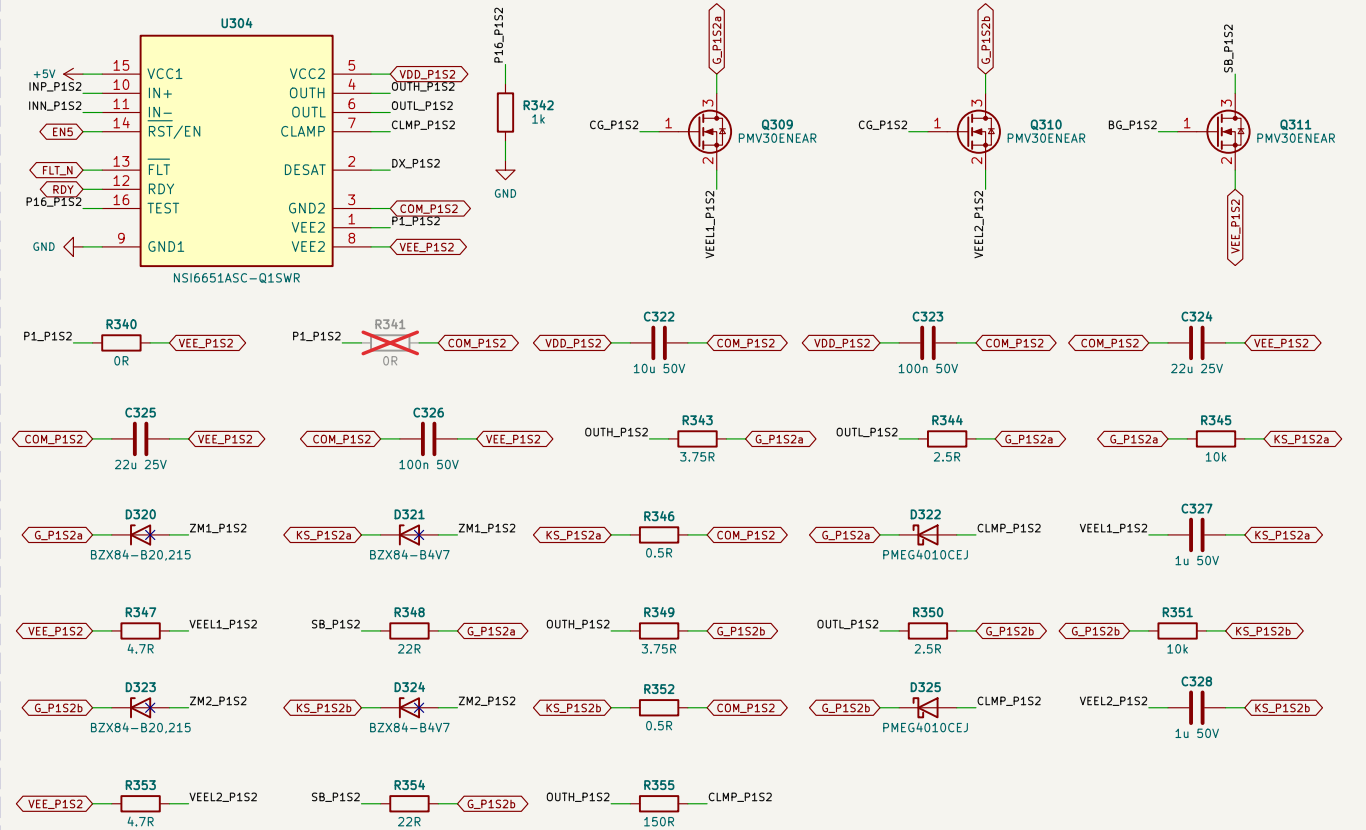
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) -> Rk - 3 x US1M - drain; BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

P1S2: LOGIC SIDE



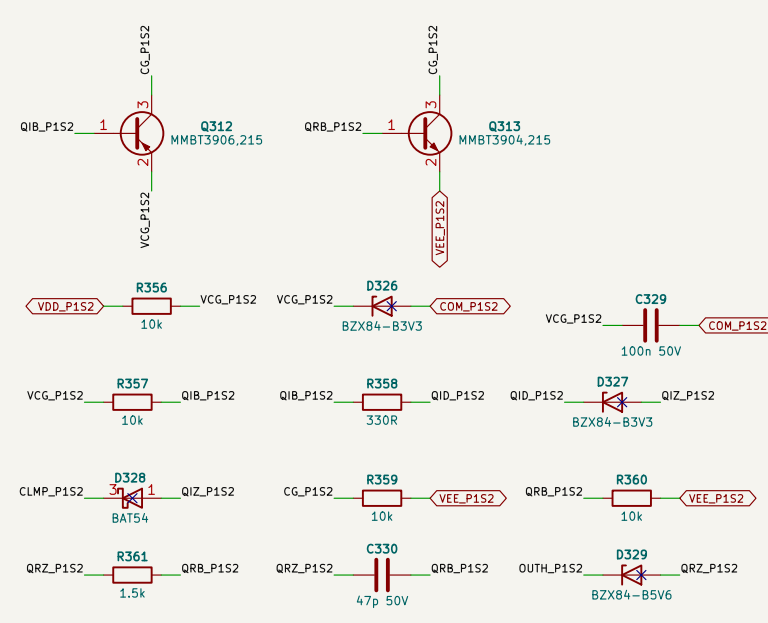
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S2: DRIVER AND 2 GATES



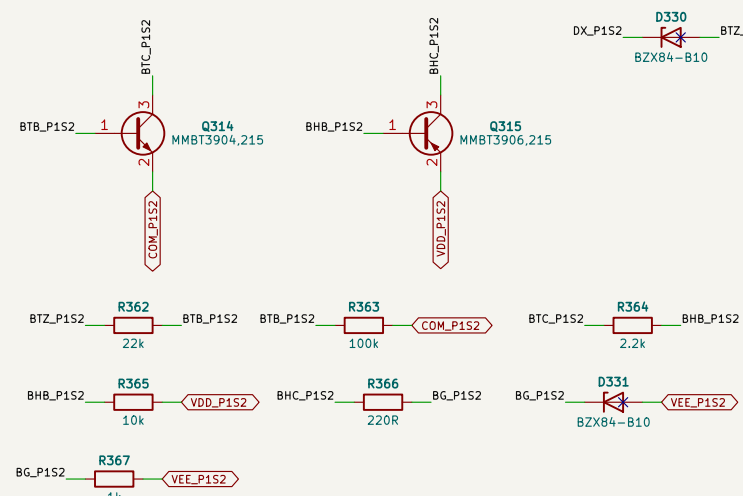
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S2: MILLER CLAMP CONTROL



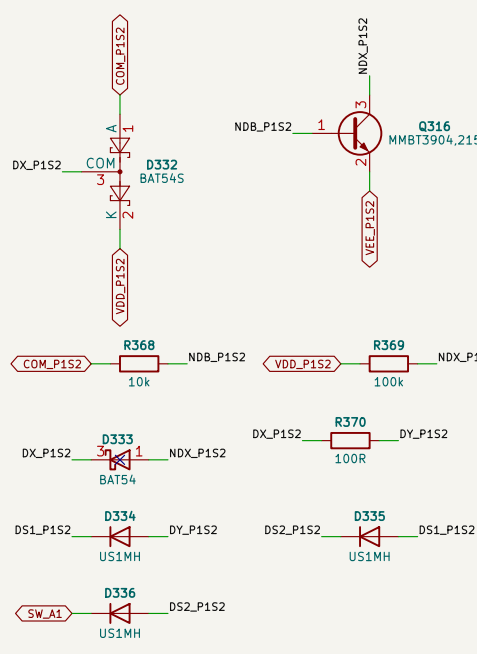
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ340DA + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P1S2: DESAT

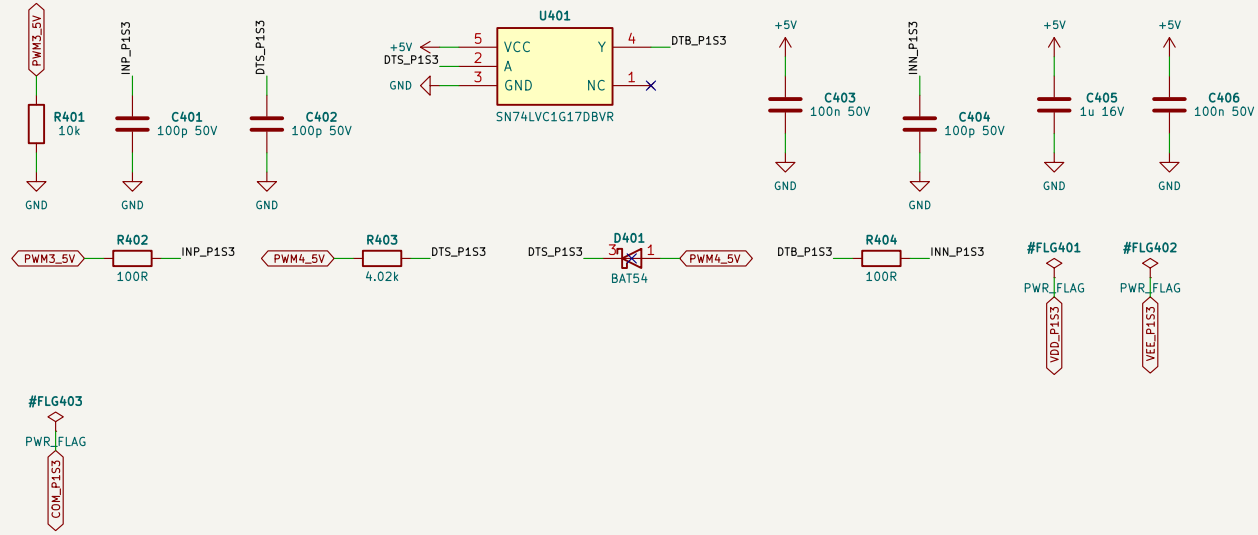


DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) -> Rk - 3 x US1M - drain; BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

LEG A: LAYOUT RULES

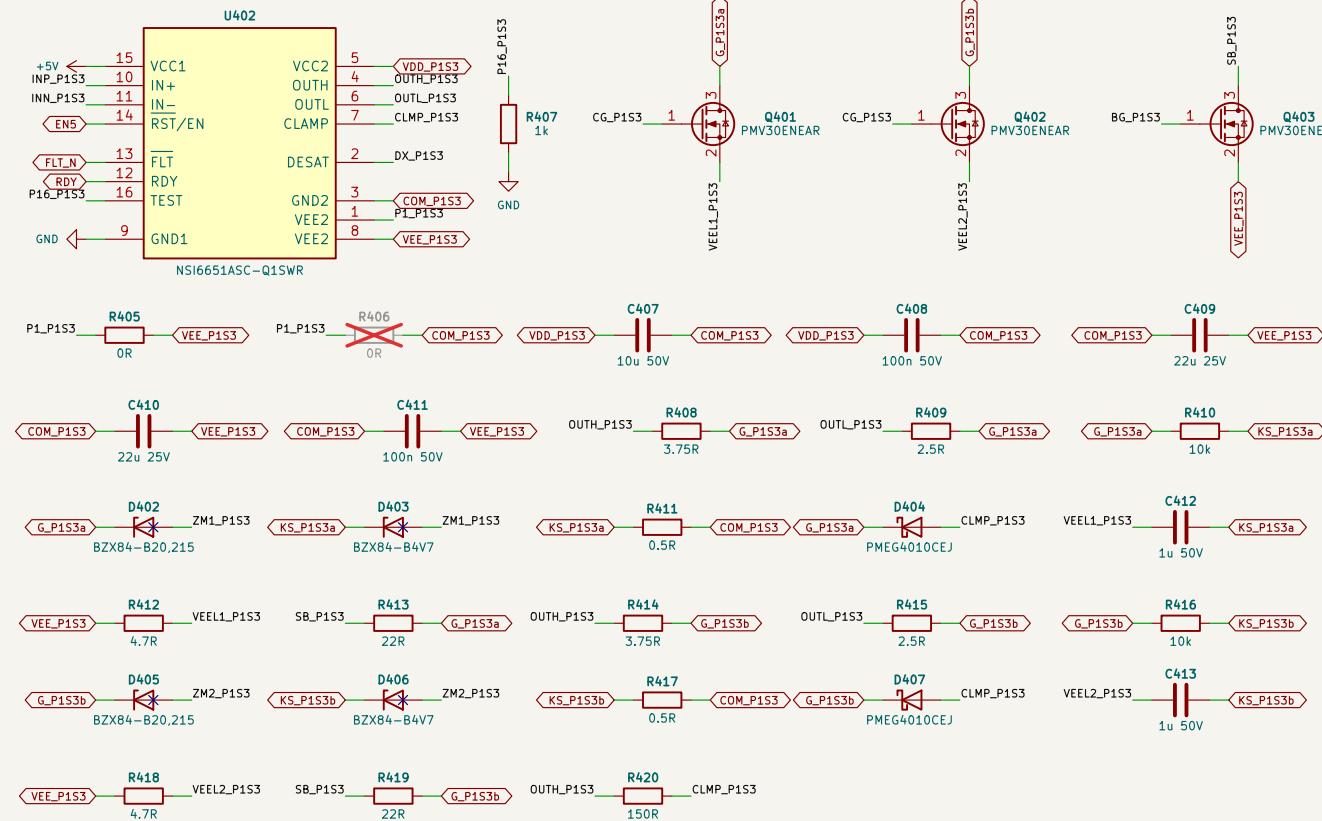
Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device; Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH; DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22); Miller hold (highside, gnd); die +1.41 V vs VGS(on) min 1.94 V at 1/8 " - BENCH TEST; Driver input side and bias primary decoupled to GND locally (CM current).

P1S3: LOGIC SIDE



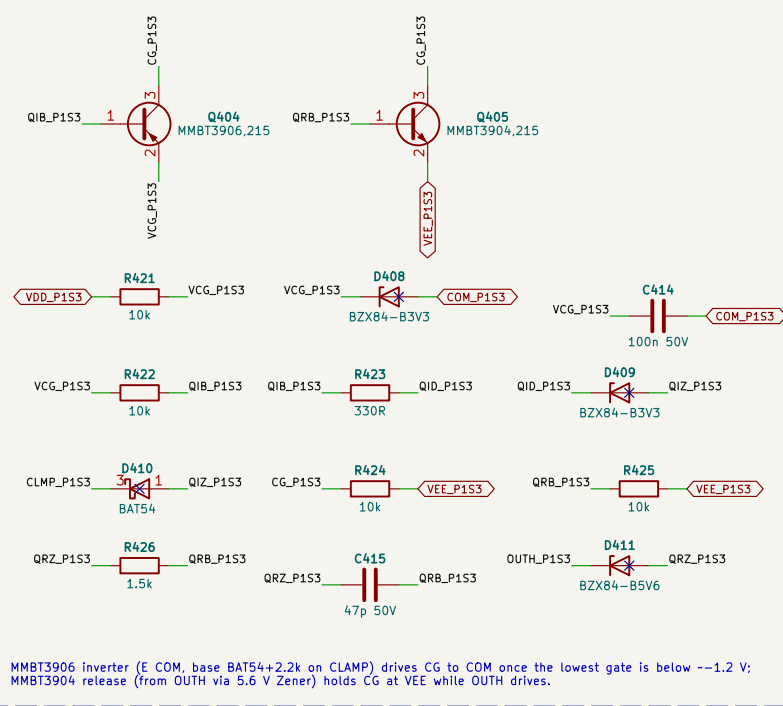
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S3: DRIVER AND 2 GATES



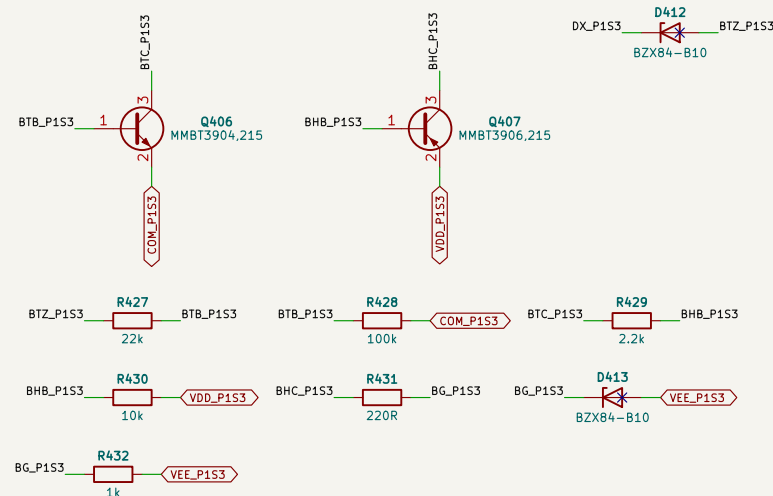
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30EENAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S3: MILLER CLAMP CONTROL



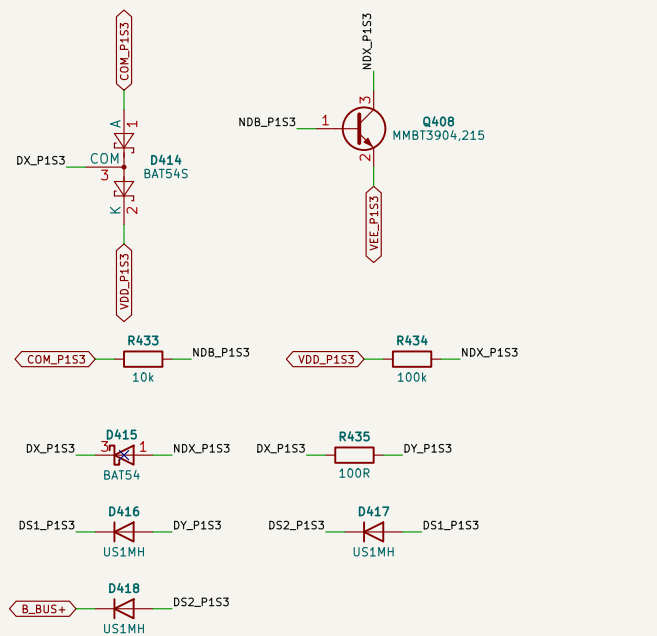
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S3: SHORT-CIRCUIT BOOSTER



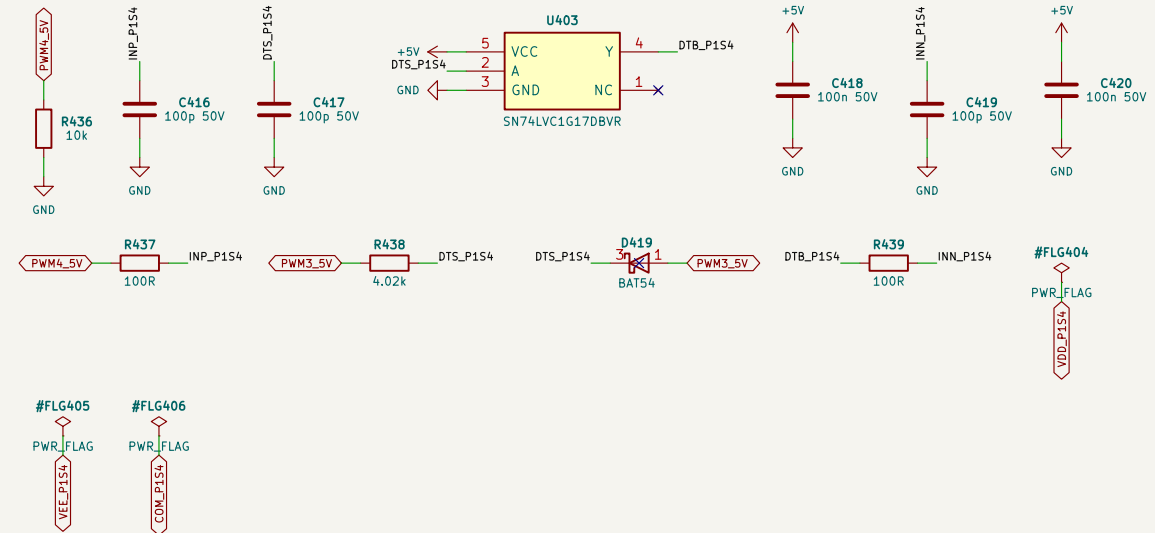
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P1S3: DESAT



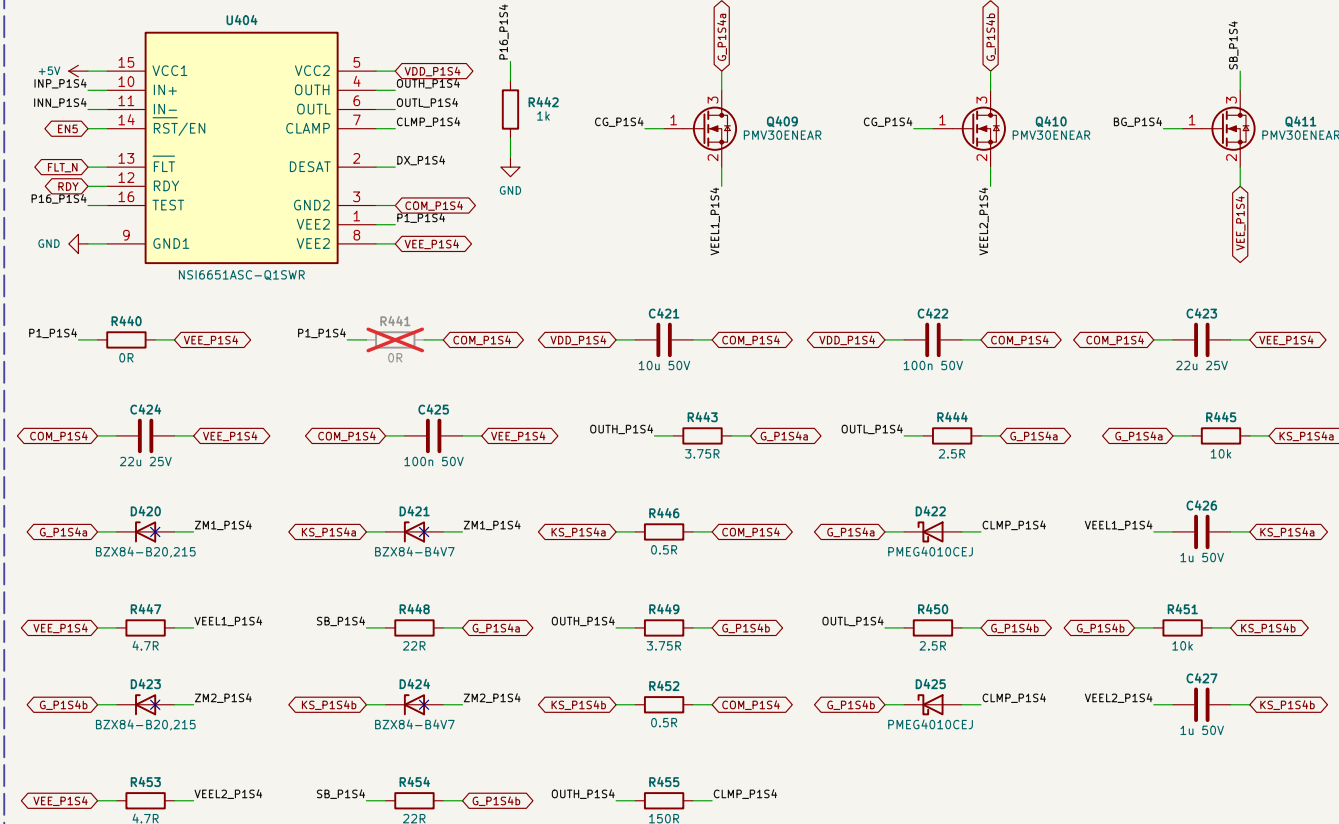
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P1S4: LOGIC SIDE



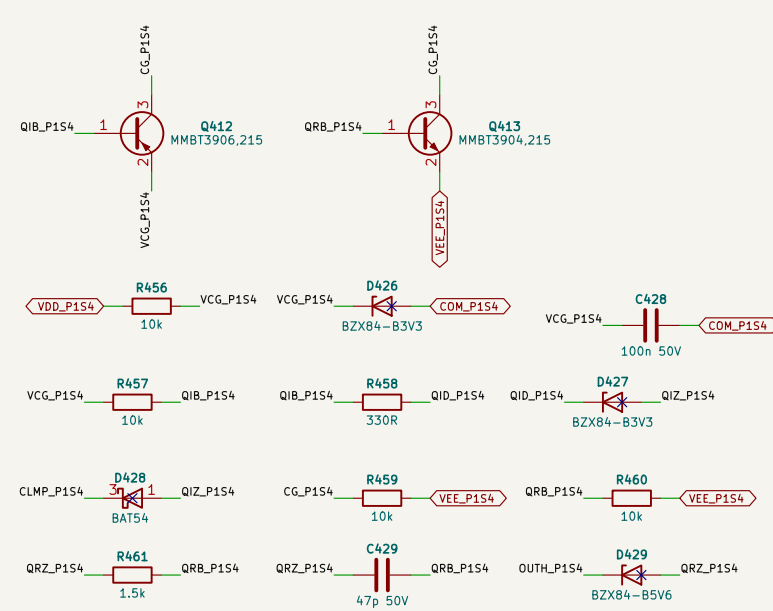
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S4: DRIVER AND 2 GATES



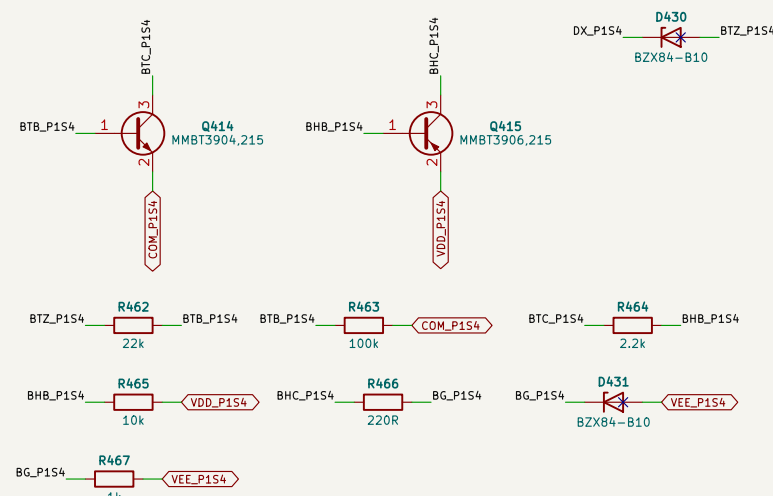
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30EENAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S4: MILLER CLAMP CONTROL



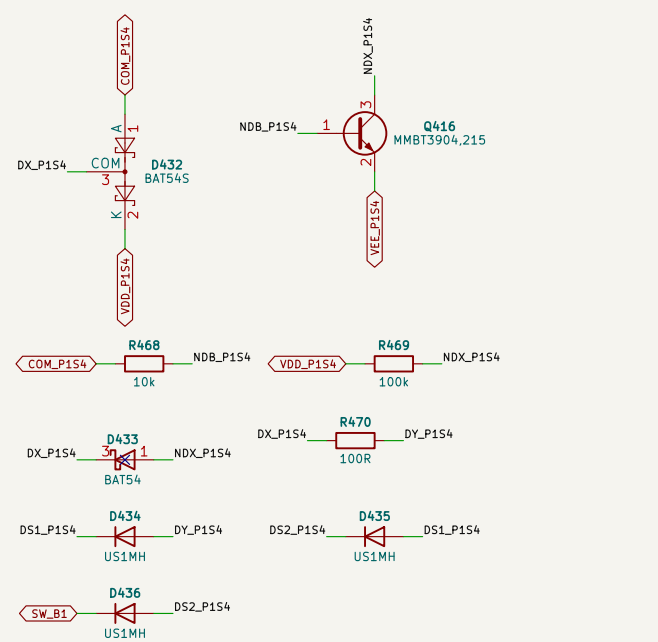
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P1S4: DESAT

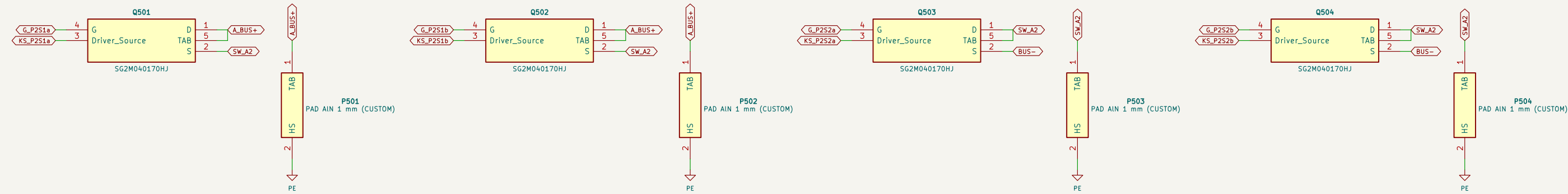


DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

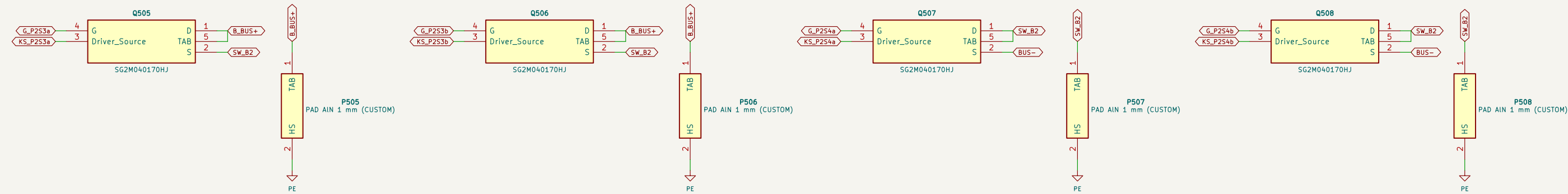
Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nspice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5 - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 2 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



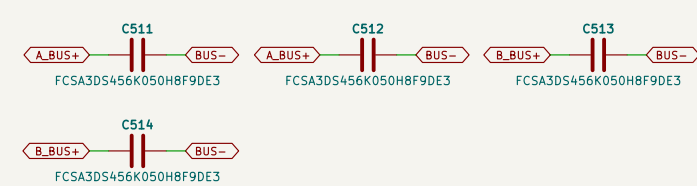
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AlN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



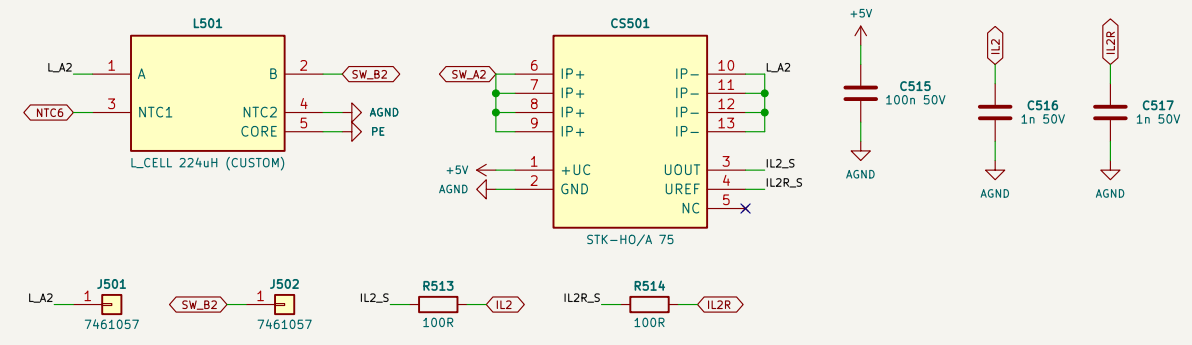
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 SHARE OF THE PORT BANKS



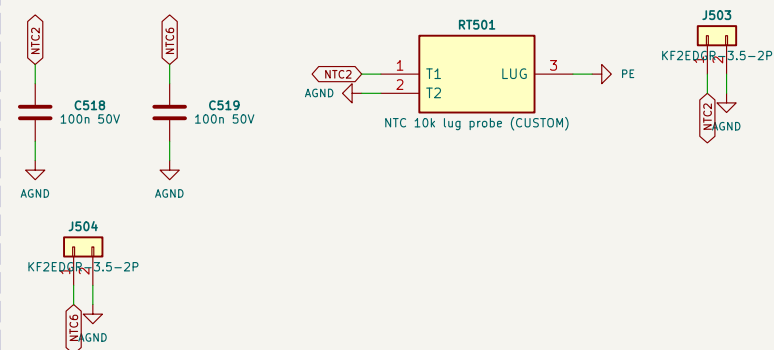
2 x FCSA3DS456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 2 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



Lead A from SW_A2 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; + = SW_A -> SW_B) to stud SW_A2; lead B on stud SW_B2. IL2 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A; 100R +/- 1 nF at the source. IL2R = the sensor's own Uref (pin 4), 100R +/- 1 nF: PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 2 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)

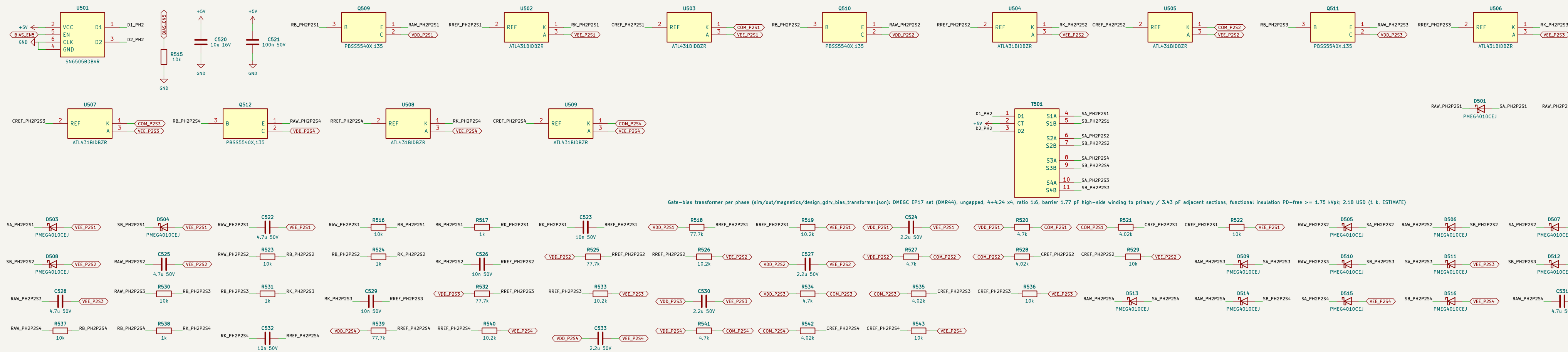


NTC2 = heatsink section of phase 2: lug probe on the EARTHED heatsink, ≥ 2.5 kV rms lead-to-lug (R-10)
NTC6 = inductor winding NTC (functional to the winding) 100 nF at the header against switch-node noise

NTC2 = heatsink section of phase 2: lug probe on the EARTHED heatsink, ≥ 2.5 kV rms lead-to-lug (R-10)
NTC6 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

NTC6 = inductor winding NTC (functional to the winding). 100 nF at the header against switch-node noise.

PH2: GATE BIAS (PER PHASE)



SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation (PD-free ≥ 1.75 kVpk, ≤ 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Chart 6

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basis

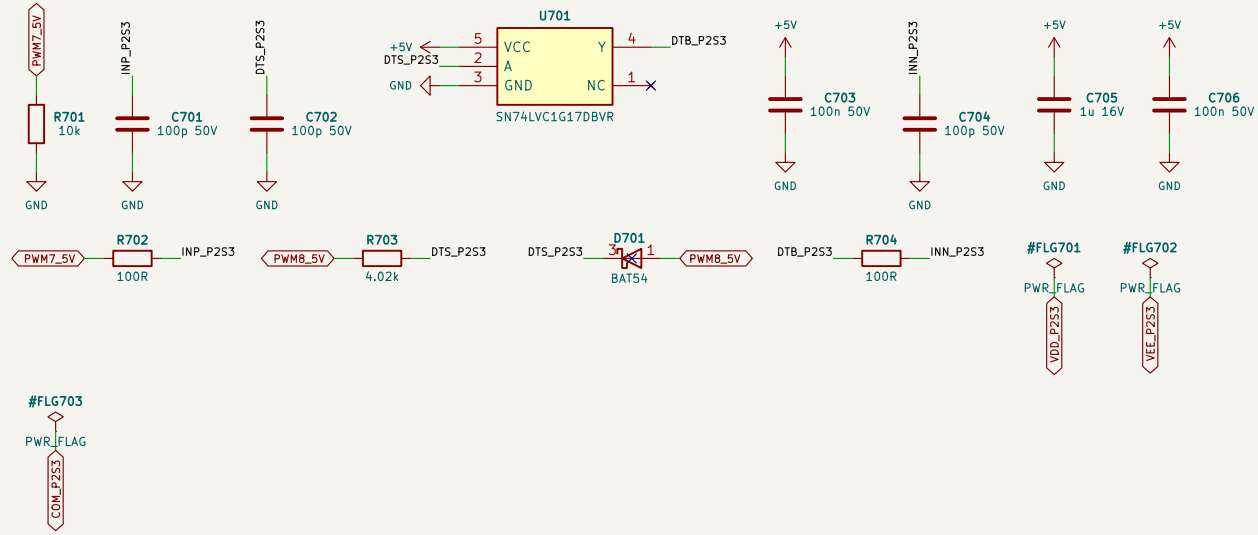
Sheet: /05 Phase 2 power stage bias sensing/

File: 05_ph2_power.kicad_sch

Title: PV-PWR-4 power board – Phase 2 power stage, bias, sensing

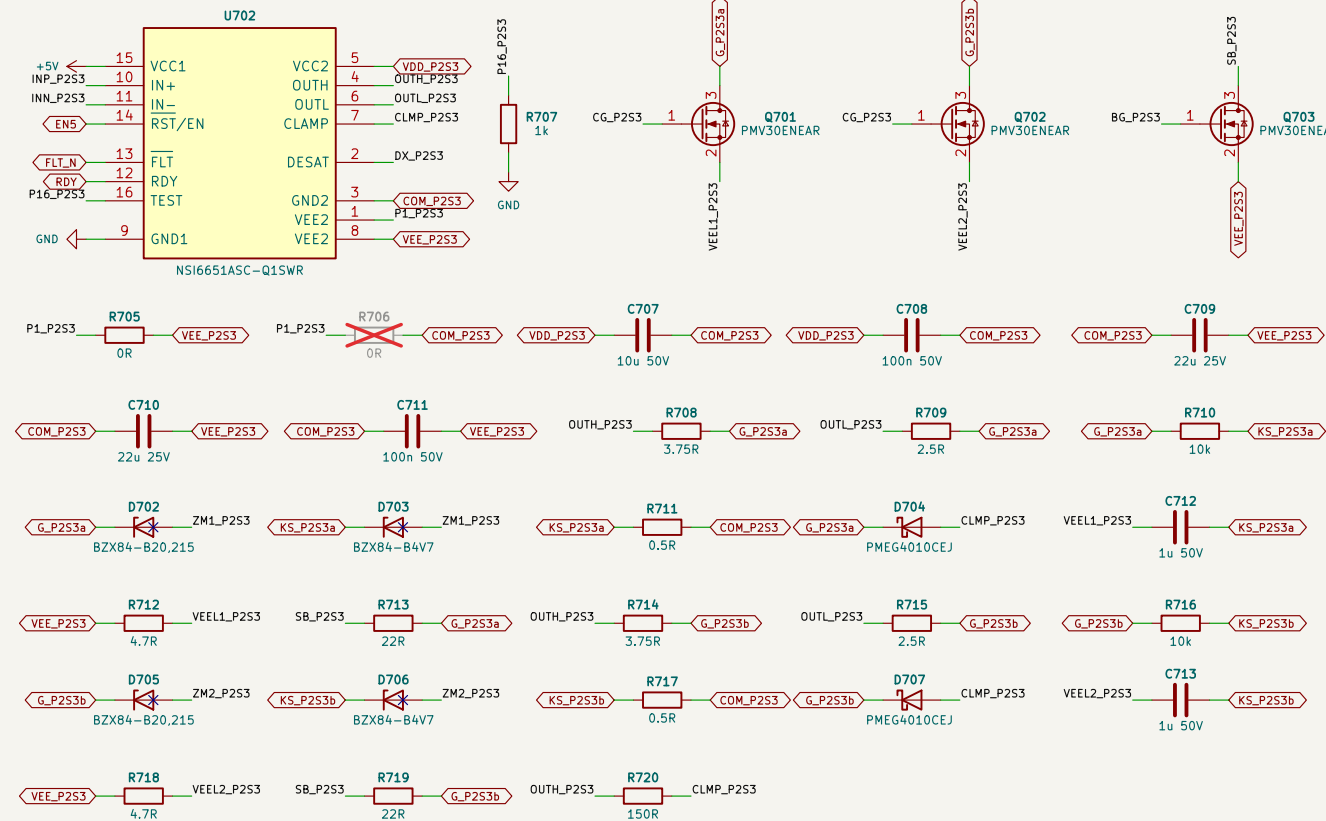
Size: A2	Date: 2026-10-05	Rev: A2
KiCad EDA 10.0.6		Id: 6/26

P2S3: LOGIC SIDE



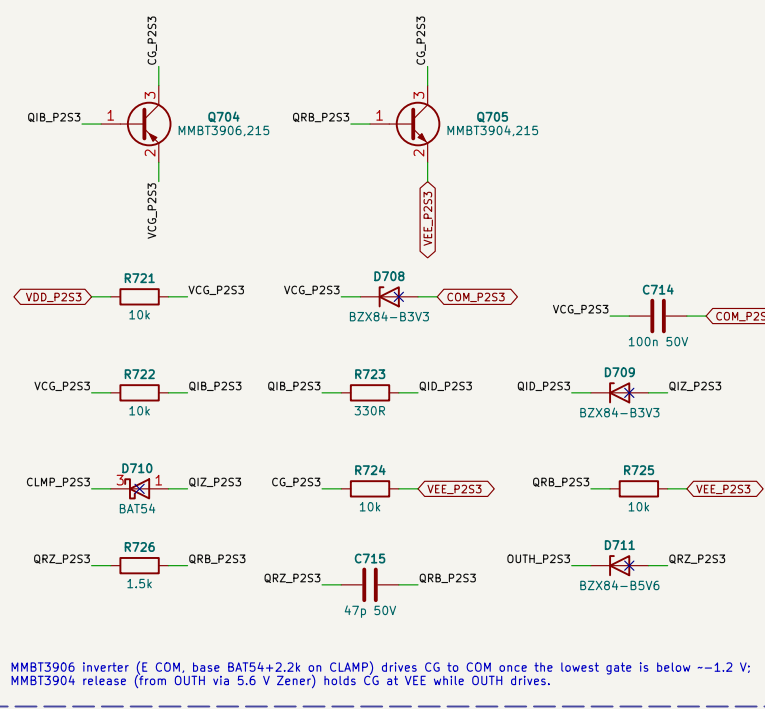
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S3: DRIVER AND 2 GATES



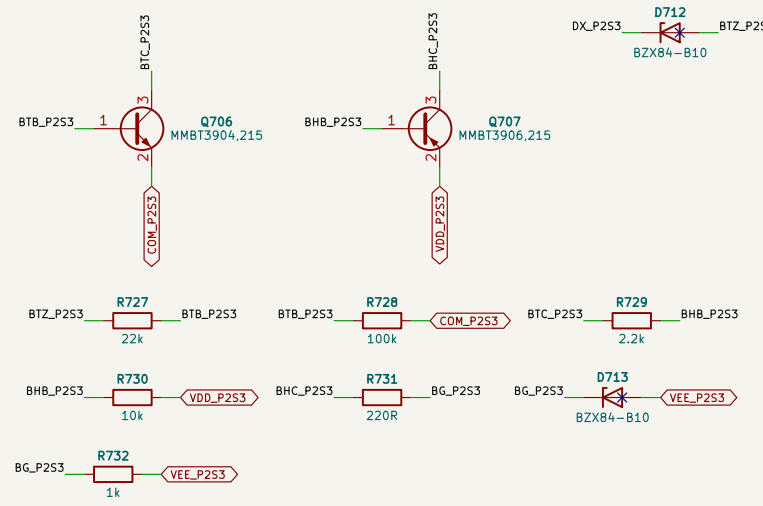
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S3: MILLER CLAMP CONTROL

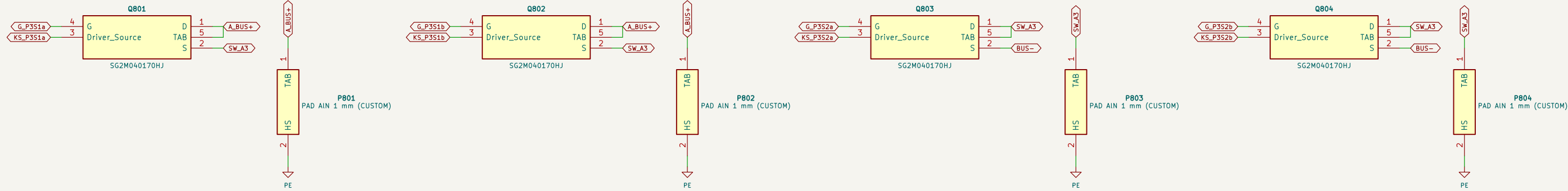


MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S3: SHORT-CIRCUIT BOOSTER

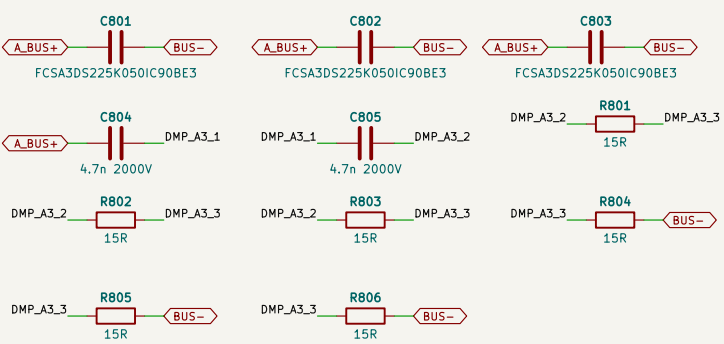


PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



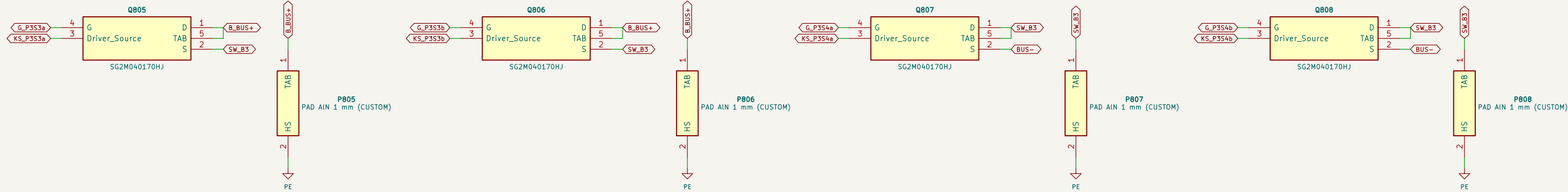
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



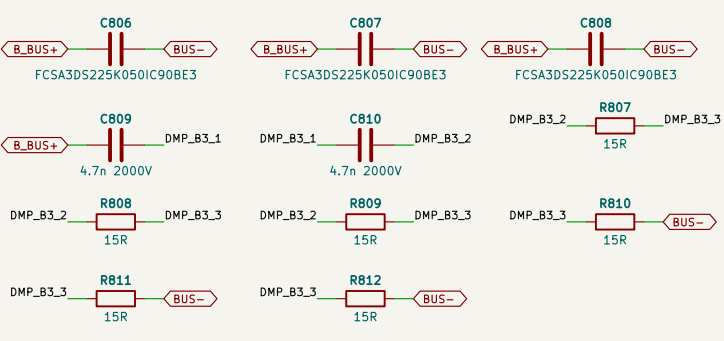
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



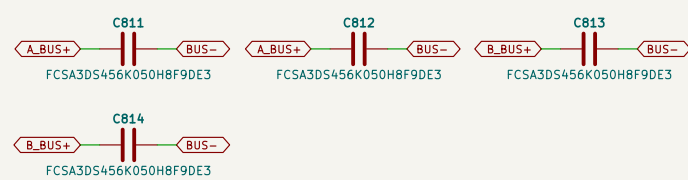
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



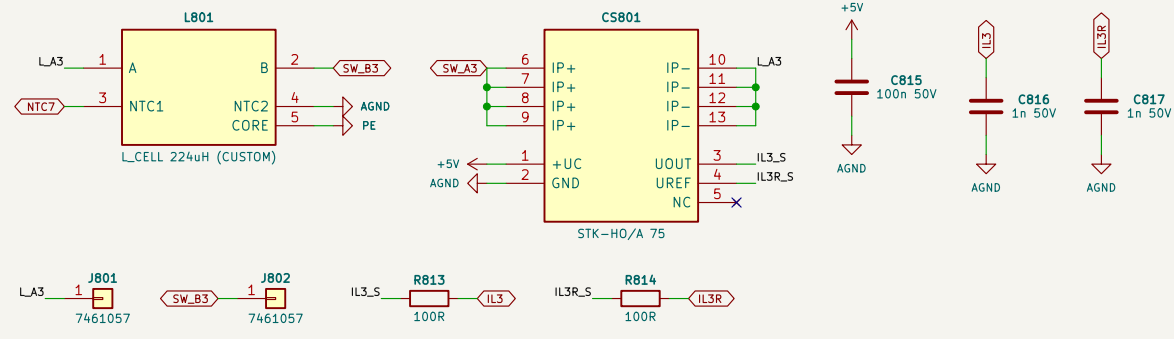
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



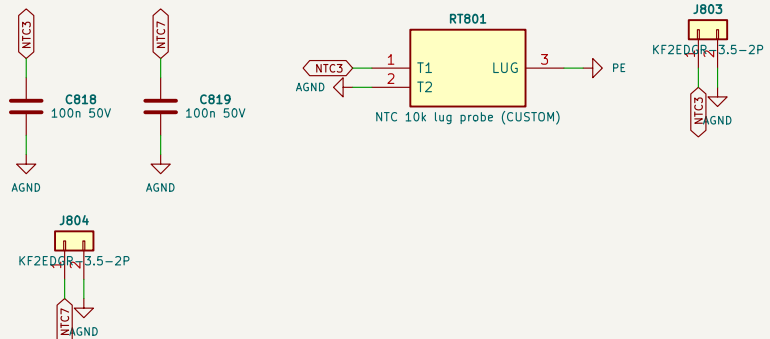
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



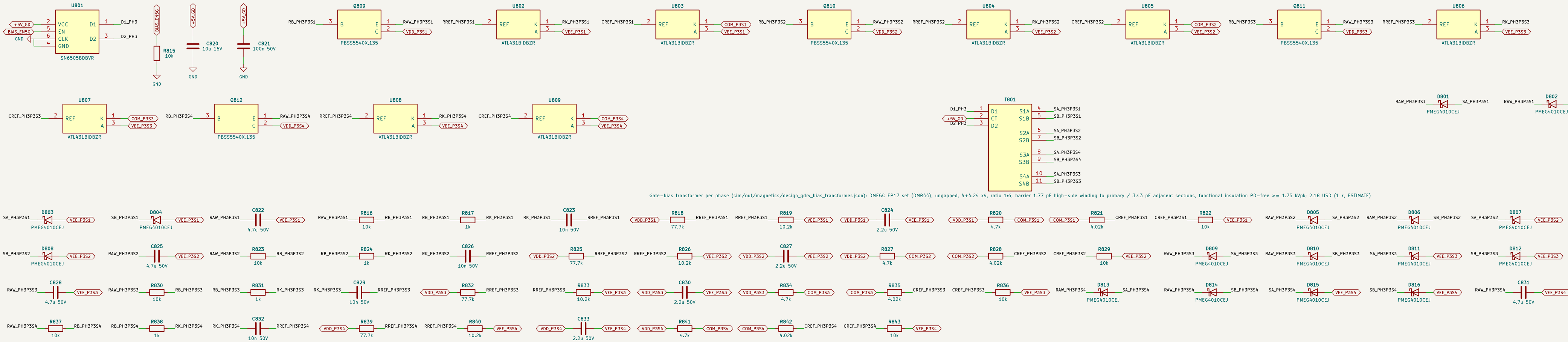
Lead A from SW_A3 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out; + = SW_A -> SW_B) to stud LA3; lead B on stud SW_B3. IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A; 100R +/- 1 mV at the source. IL3R = the sensor's own Uref (pin 4). 100R +/- 1 mV: PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



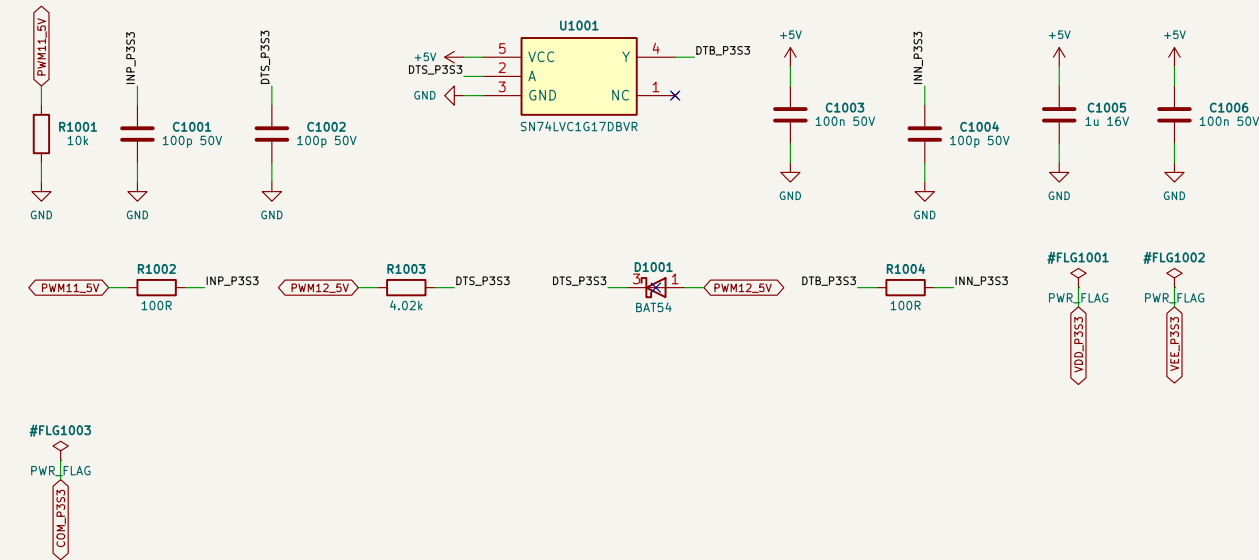
NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10). NTC7 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



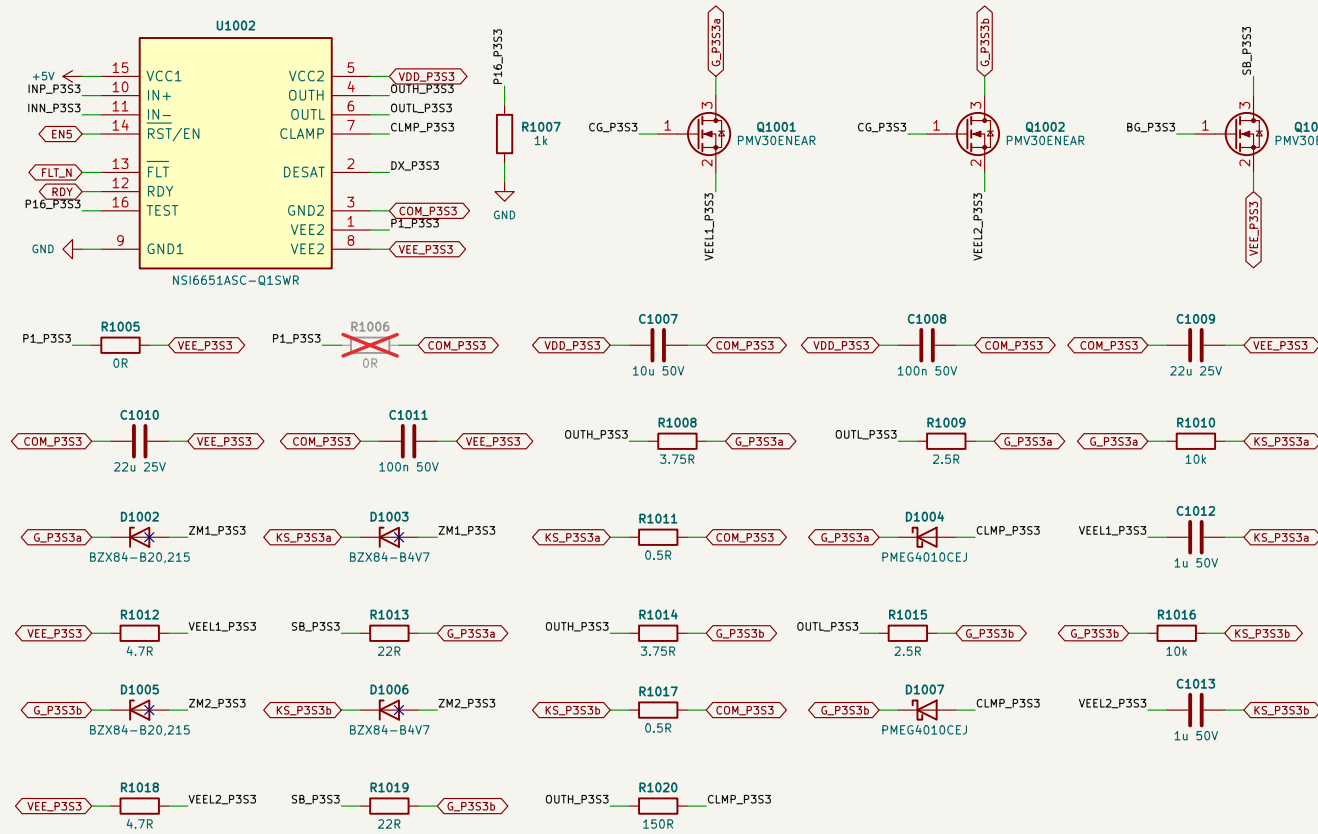
SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T_BIAS4 custom, functional insulation (PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S3: LOGIC SIDE



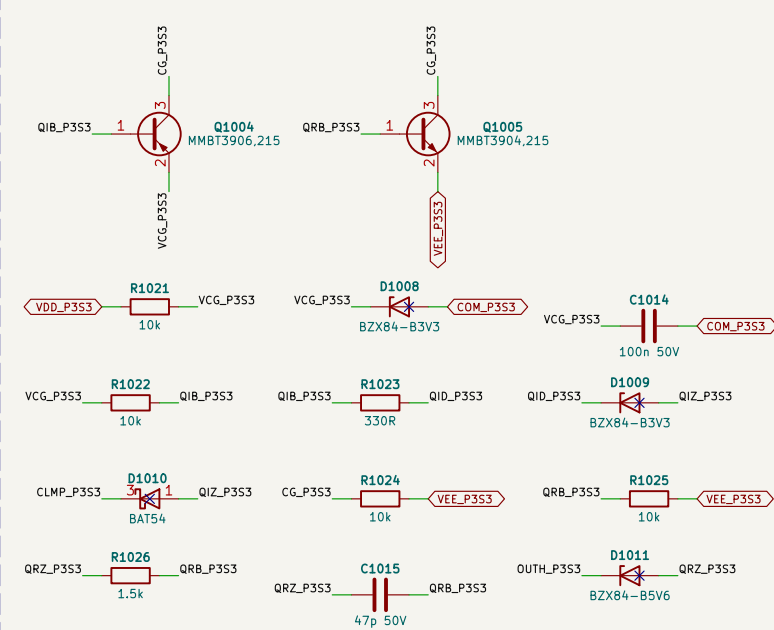
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S3: DRIVER AND 2 GATES



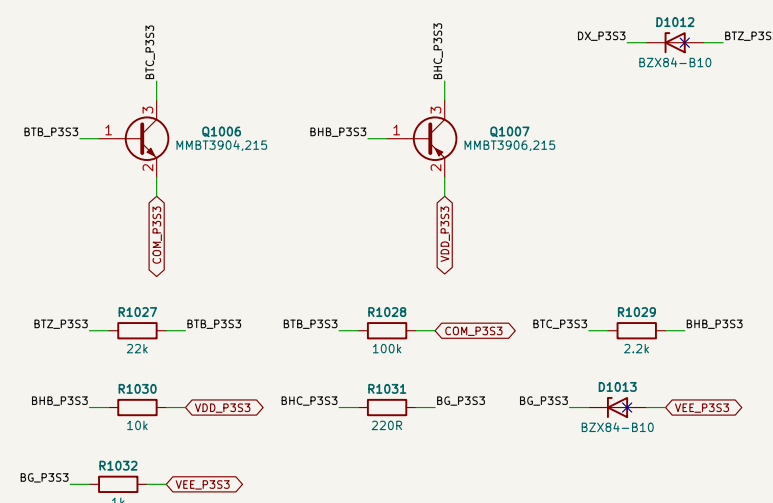
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S3: MILLER CLAMP CONTROL



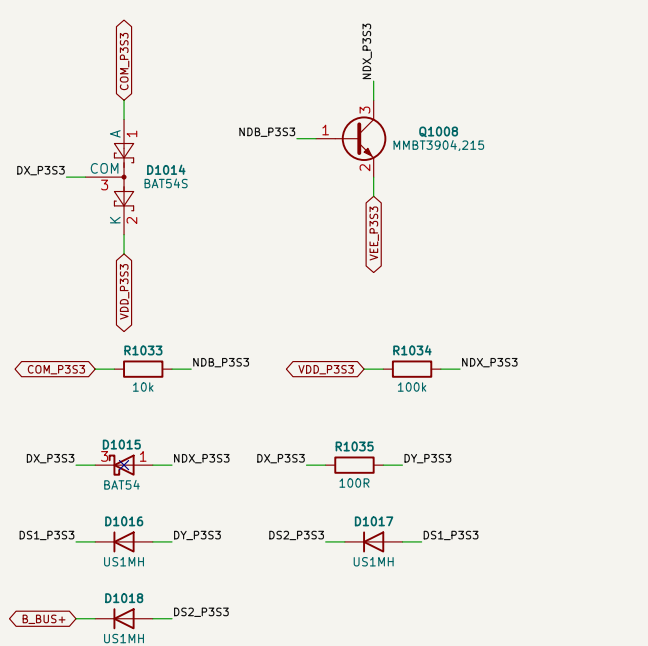
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below ~ -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S3: SHORT-CIRCUIT BOOSTER



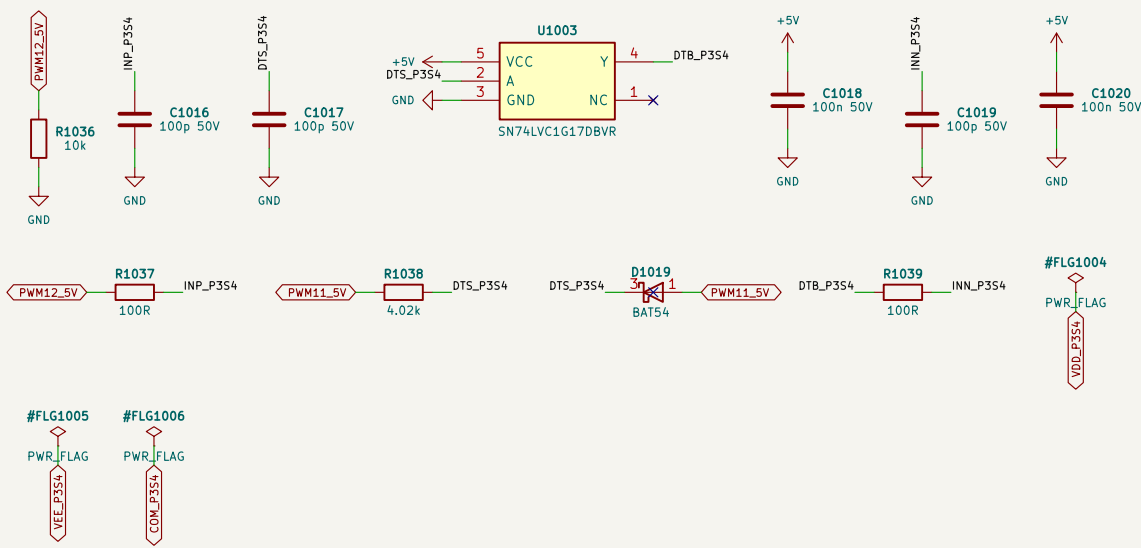
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S3: DESAT



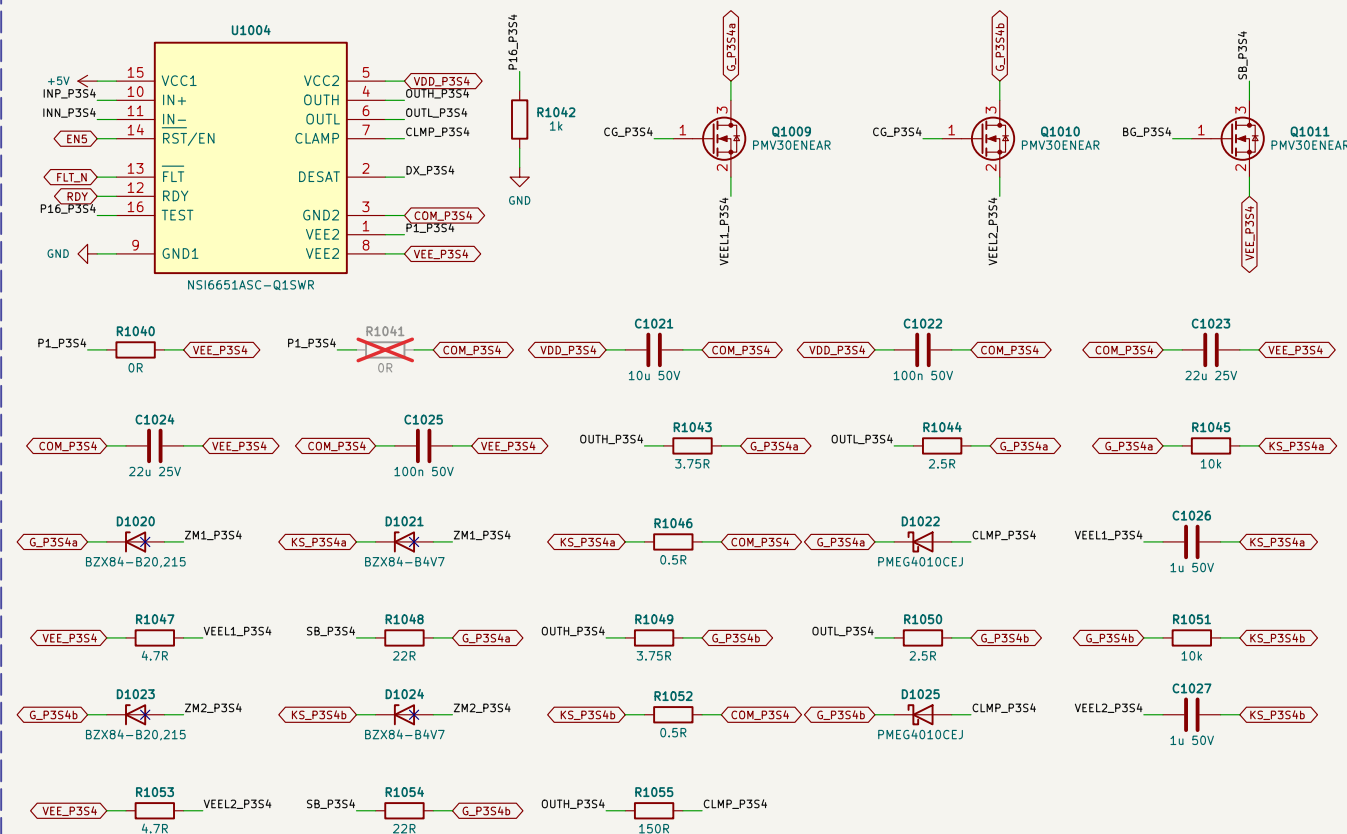
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P3S4: LOGIC SIDE



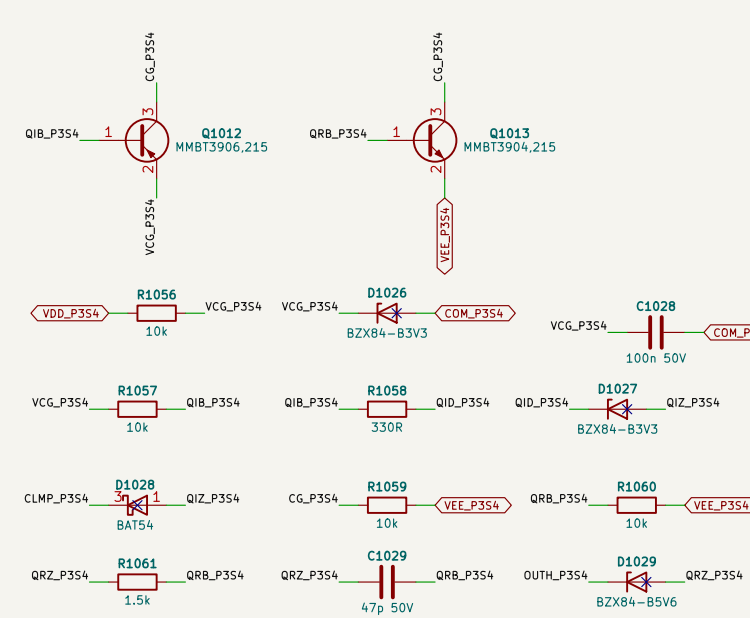
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S4: DRIVER AND 2 GATES



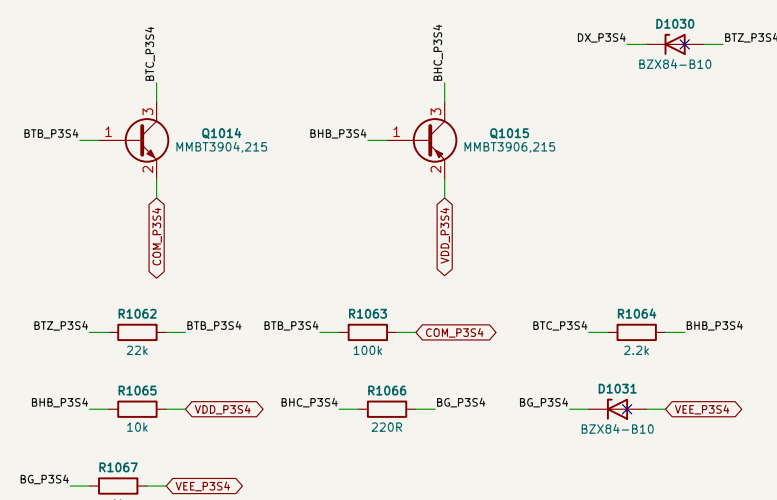
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S4: MILLER CLAMP CONTROL



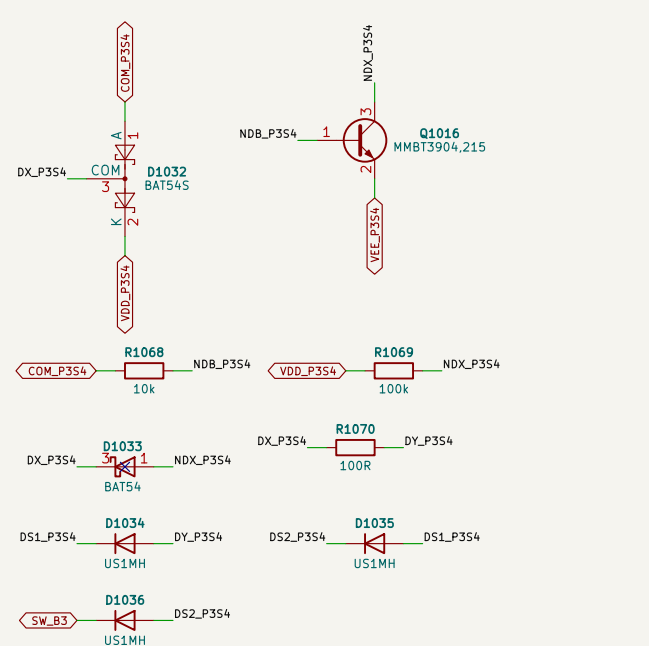
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 14

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE base

Sheet: /10 Phase 3 leg B gate drive (S3, S4)/

File: 10_ph3_gdb.kicad_sch

Title: PV-PWR-4 power board - Phase 3 leg B gate drive (S3, S4)

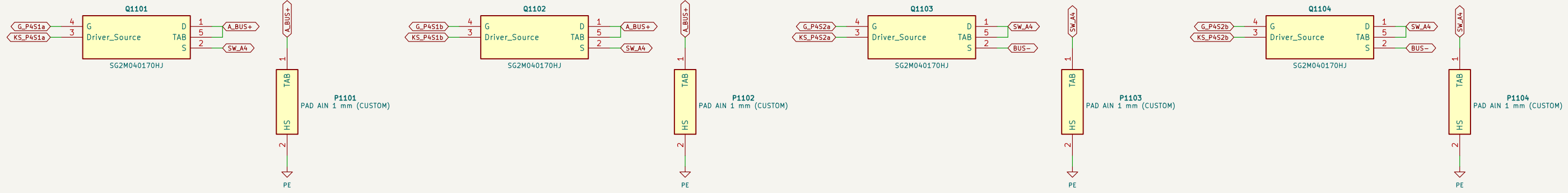
Size: A2	Date: 2026-10-05
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Rev: A2

KiCad E.D.A. 10.0.6

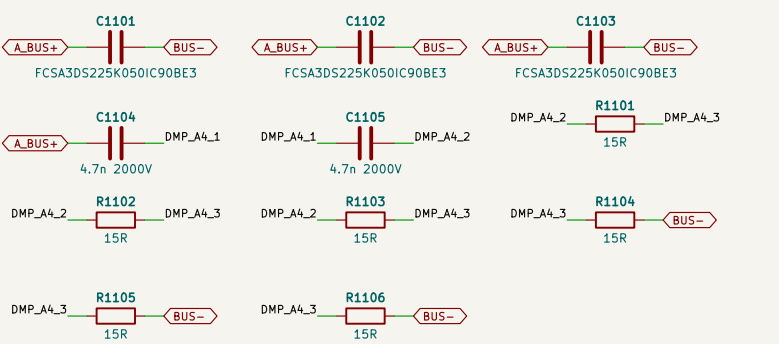
Id: 11/20

PHASE 4 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



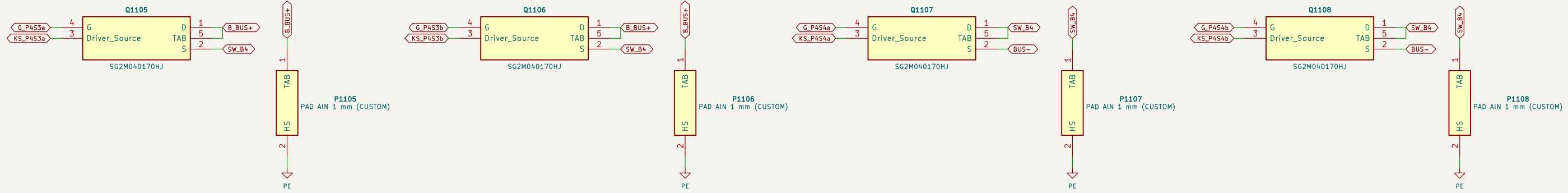
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG A DECOUPLING + RC DAMPER



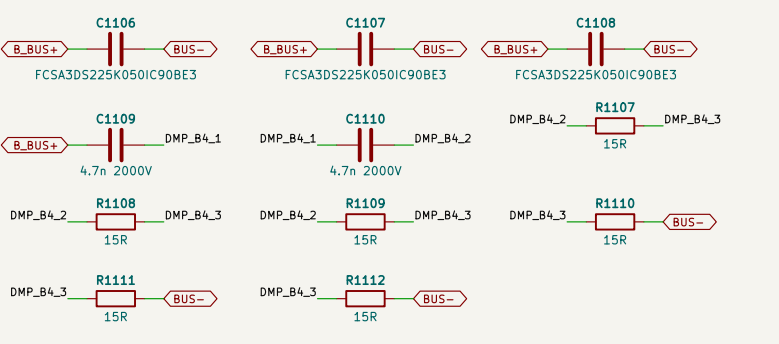
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 4 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



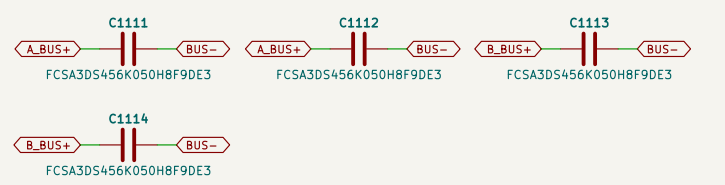
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG B DECOUPLING + RC DAMPER



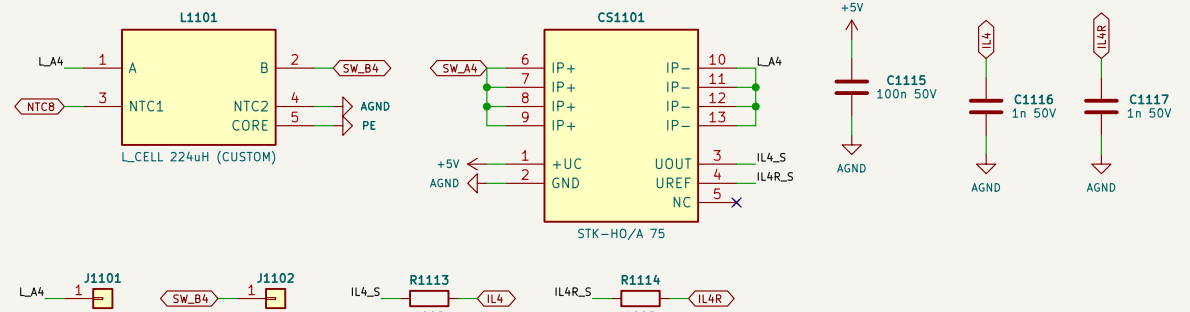
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 4 SHARE OF THE PORT BANKS



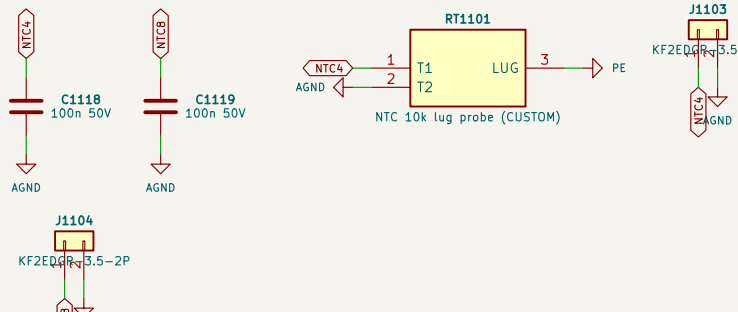
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 4 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



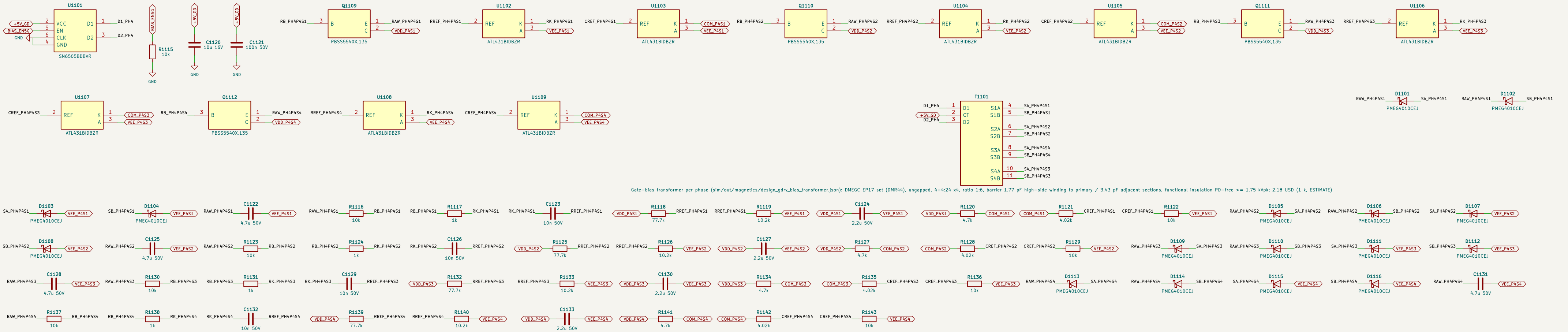
Lead A from SW_A4 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out) + = SW_A -> SW_B to stud
L_A4; lead B on stud SW_B4; IL4 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mV at the source. IL4R = the sensor's own Uref (pin 4). 100R +/- 1 mV; PV-CTL references its trip
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 4 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC4 = heatsink section of phase 4: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC5 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH4: GATE BIAS (PER PHASE)



PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{JN}- / A_{JN}-, bank A_{BUS}-. No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec_lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN}+ / A_{IN}-, bank A_{BUS}+. No fuse, no precharge; array lsc ≤ 1.25 x rating.
Contactor opens ≤ 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

TYT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up. eff 3793 V pole-P1 and pole-pole at in 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

Sheet: /14 Port A power path (pv)/		
File: 14_leanA_power.kicad_sch		
Title: PV-PWR-4 power board - Port A power path (pv)		
Size: A2	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 15/26

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 16
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Port A sensing and interlocks/
File: 15_leanA_ctrl.kicad_sch

Title: **PV-PWR-4 power board – Port A sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 16/26

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 17
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Port A coil drives/
File: 16_leanA_drive.kicad_sch

Title: **PV-PWR-4 power board – Port A coil drives**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 17/26

Terminal side B_{LN}+ / B_{LN}-, bank B_{BUS}+, HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side B_{LN}+ / B_{LN}-, bank B_{BUS}+, HIPE501 250 A wR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

[illegible]

TV125751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE-T, U_{eff} 3793 V pole-PE and pole-pole at 3 kA (calc. = 4.0 kV) => reinforced 6 kV / basic 4 kV, D=032; Imax 25 kA, Branch fuses ITI CH145 50kV 36 A (30 kA / I₂ t₂ ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

Not bench-validated. Values read from sim/out/*_json; checks: design_check() Sheet 18 Schematics authored by Chinmoy Bhuyan Live board referenced to bus-4 gate islands functional: SELV winding reinforced; HW-PE base Sheet: /16 Port B power path (battery)/ File: 16_cleanB_powerKick.auc			
Title: PV-PWR-4 power board - Port B power path (battery)			
Size: A1	Date: 2026-10-05	Rev: A2	
KicAd F.D.A. 10.0.6			Id: 18/26
15		16	

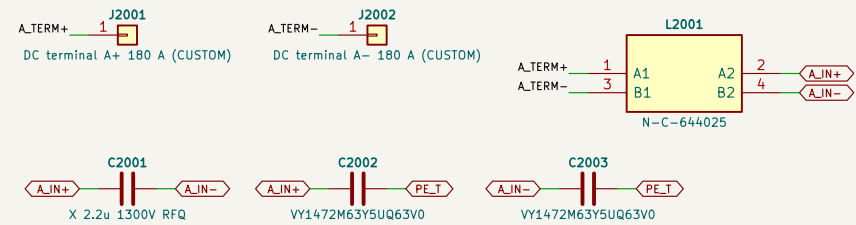
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 19 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /17 Port B sensing and interlocks/ File: 17_leanB_ctrl.kicad_sch		
Title: PV-PWR-4 power board – Port B sensing and interlocks		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 19/26	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

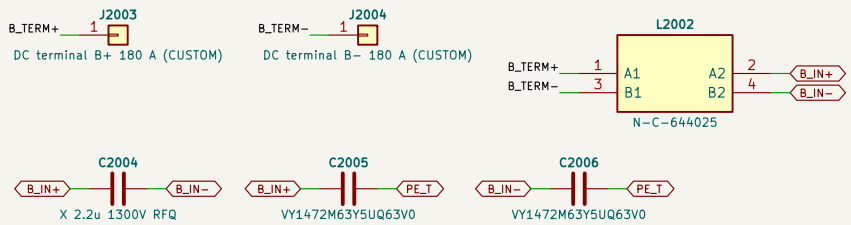
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 20 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /18 Port B coil drives/ File: 18_leanB_drive.kicad_sch		
Title: PV-PWR-4 power board – Port B coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 20/26	

PORT A TERMINALS AND EMI FILTER



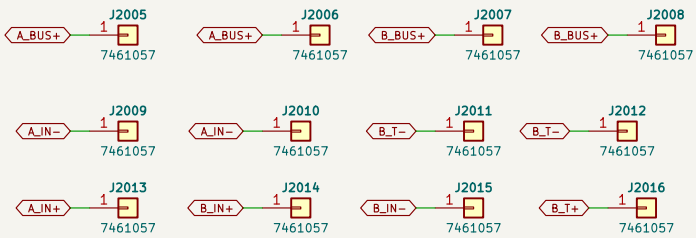
PV: chassis terminals A_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1). X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



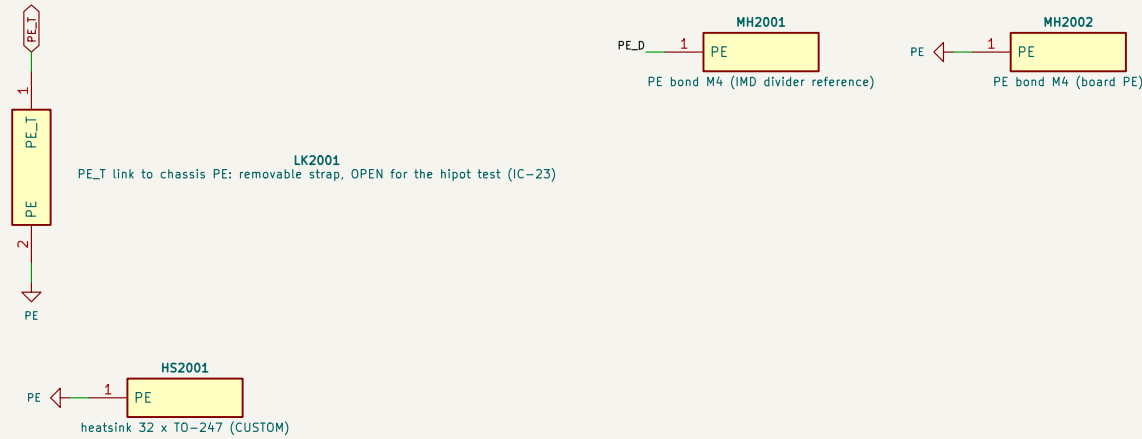
battery: chassis terminals B_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1). X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



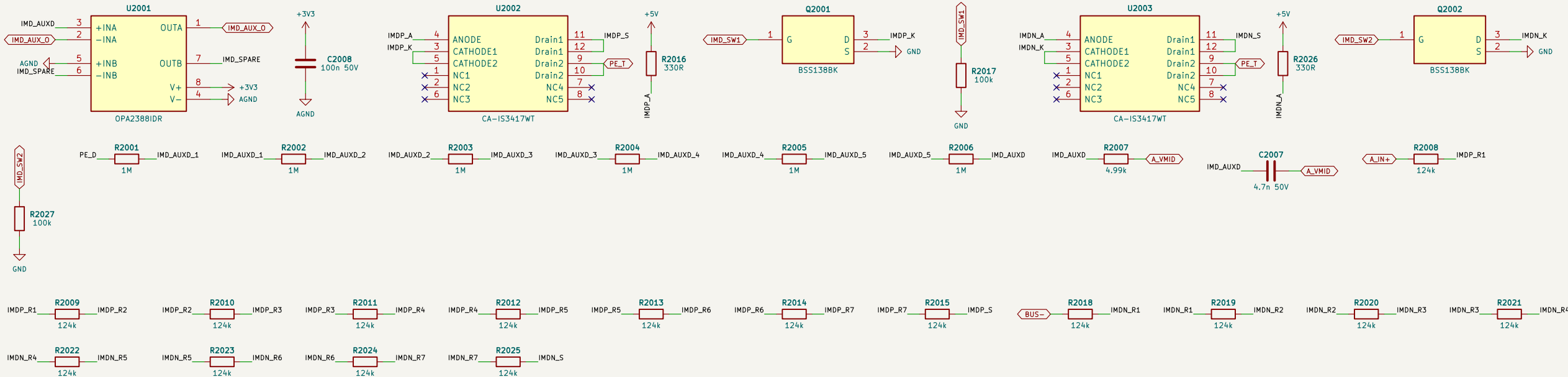
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



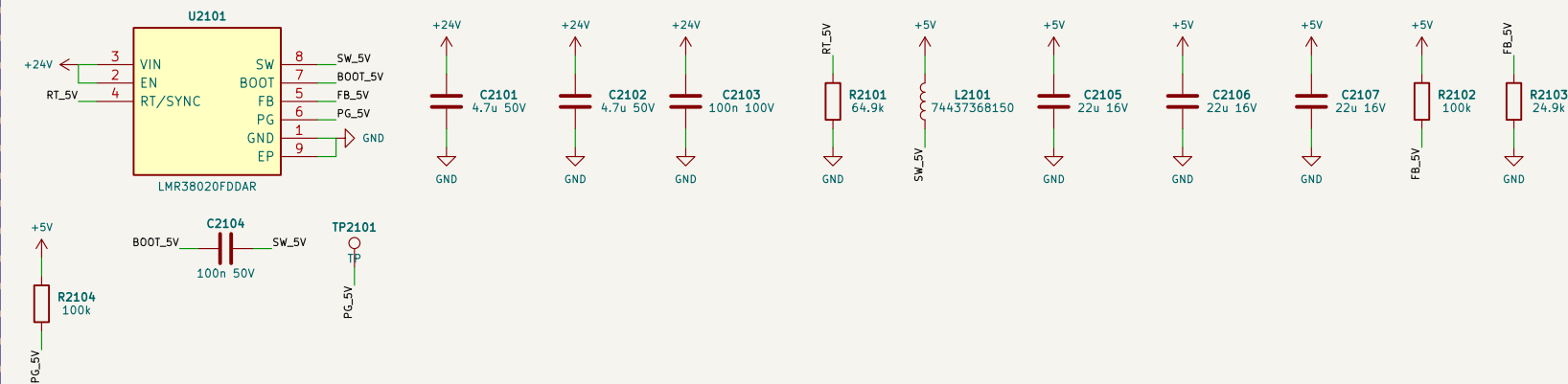
PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



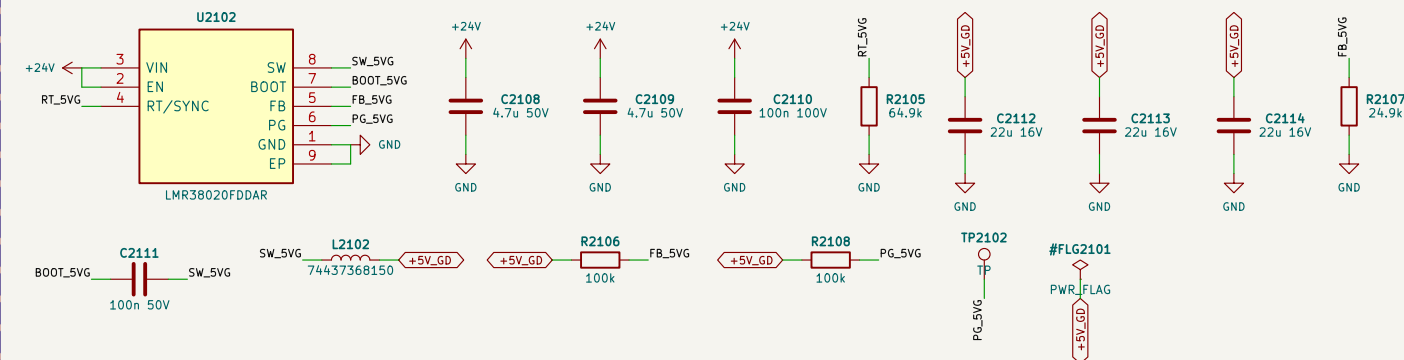
IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected); VPE = PE divider against BUS- (VMID of port A).

24 V $\rightarrow$ +5V (LMR38020, 400 KHZ, 2 A)



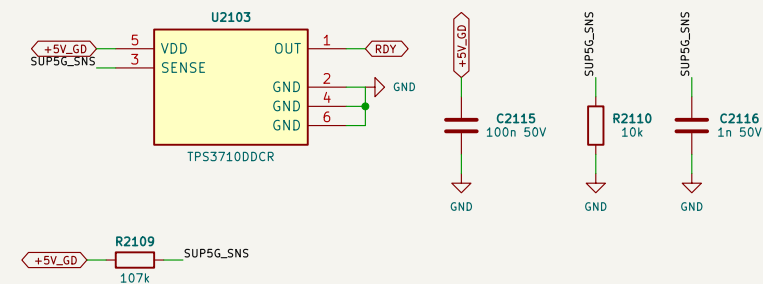
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2. ctrl_c2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



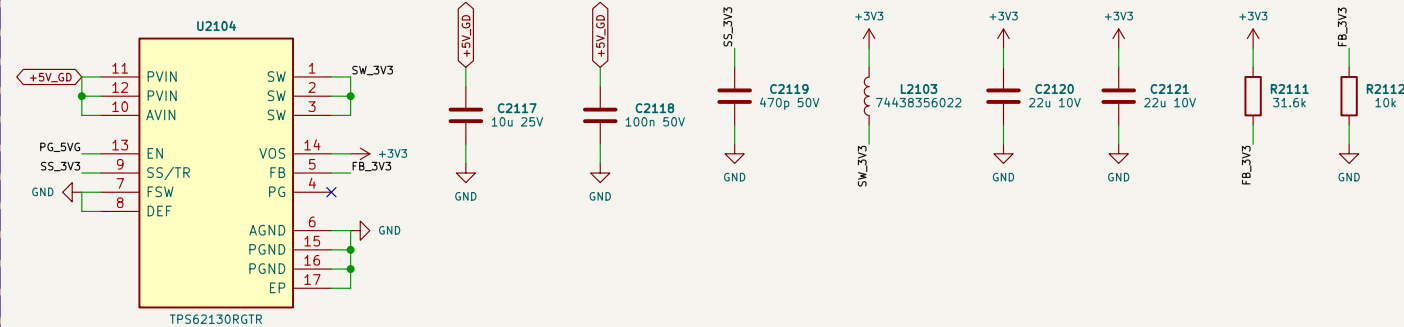
gate bias of phases 3-4. ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



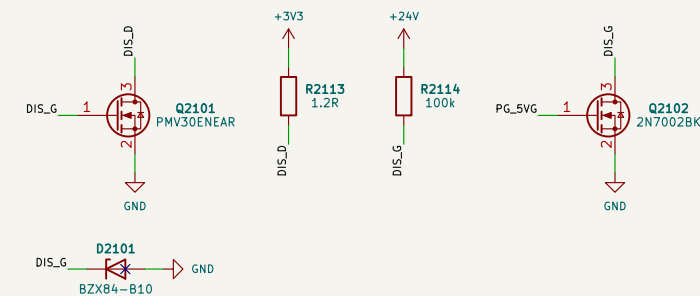
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3-4 is below 4.52-4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5V0, FB 31.6k / 10k 0.1 % $\rightarrow$ 3.240–3.433 V ($\geq$ 3.20 V for the control board's reference), SS/TR 470p C0G; ramp-up 12–16 mV/us (F28003x VDDIO 8–100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control $\leq$ 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at ≥ 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3-29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12-36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 22

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /19 Live 5 V / 3.3 V, SELV connector/

File: 19_supplies.kicad_sch

Title: PV-PWR-4 power board – Live 5 V / 3.3 V, SELV connector

Size: A3

Date: 2026-10-05

Rev: A2

KiCad E.D.A. 10.0.6

Id: 22/26

TAP A_IN+



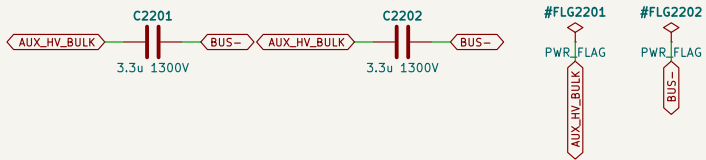
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP B_T+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual -> 1199 V (UR 1300 V); filter corner 546 Hz.

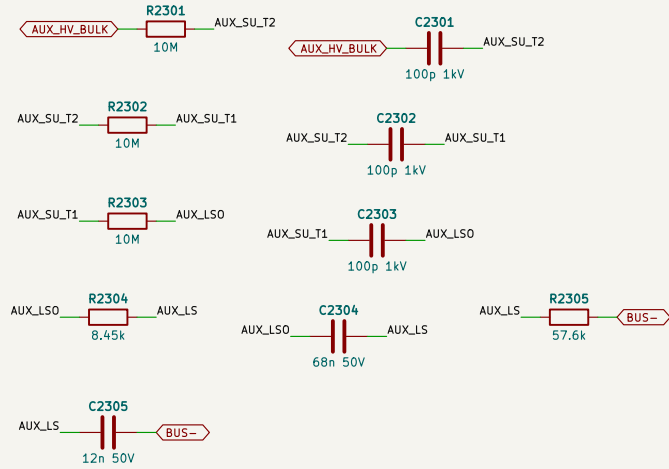
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 23
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /20 AUX75 input taps, OR, film/
File: 20_aux75_input.kicad_sch

Title: PV-PWR-4 power board - AUX75 input taps, OR, film

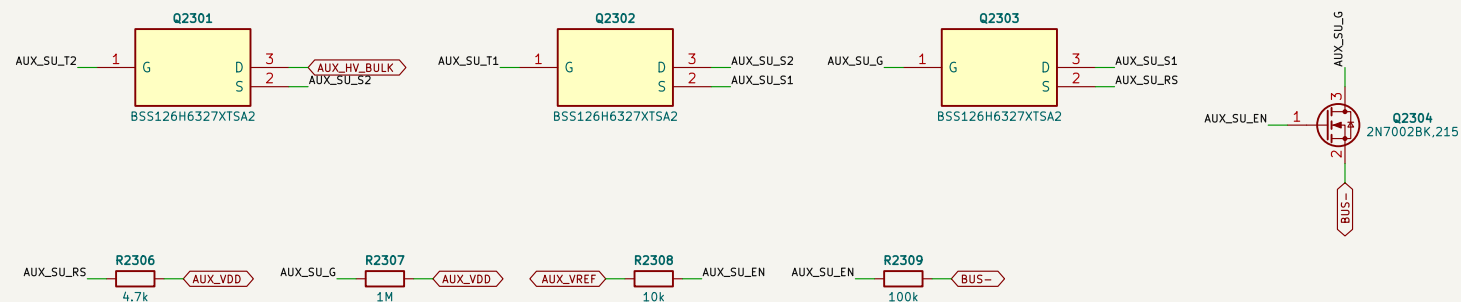
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 23/26

COMPENSATED TAP STRING



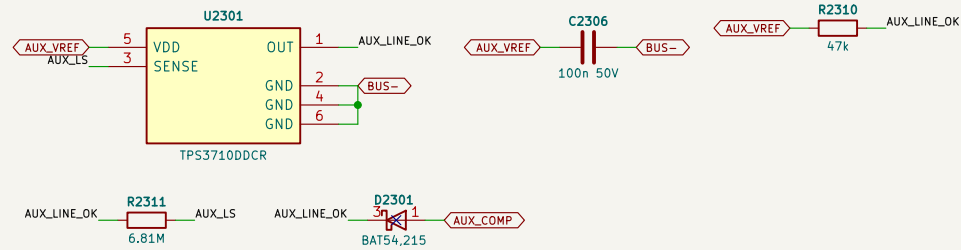
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



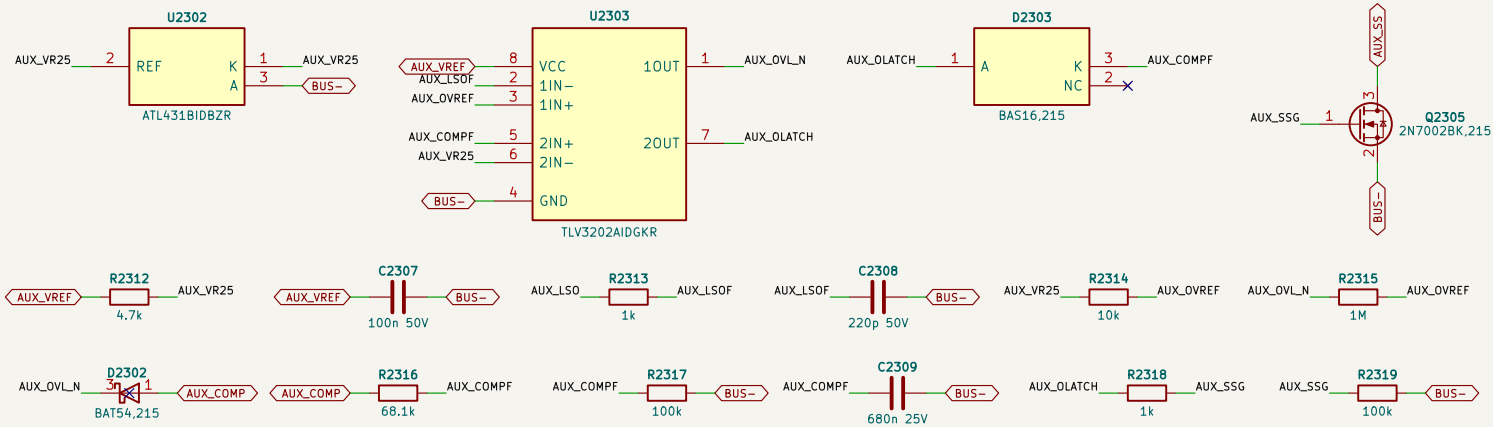
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



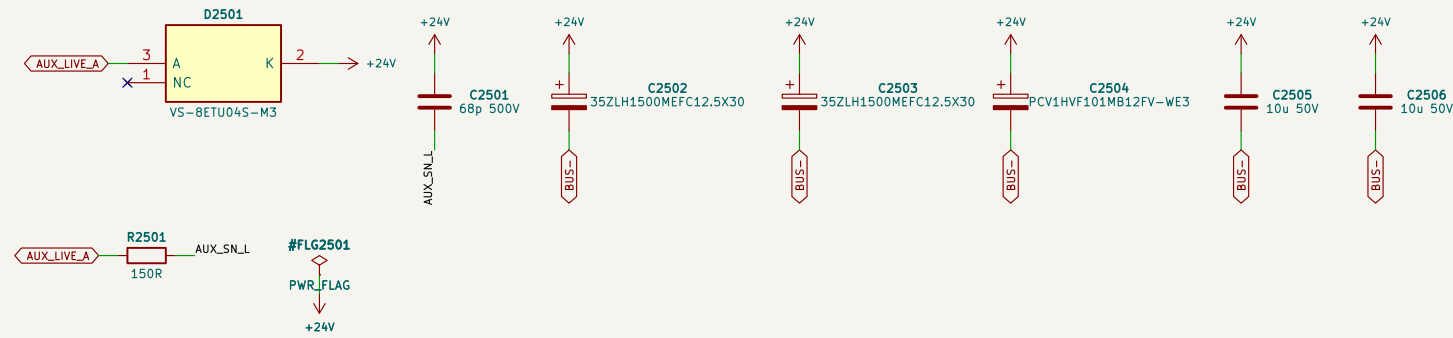
LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



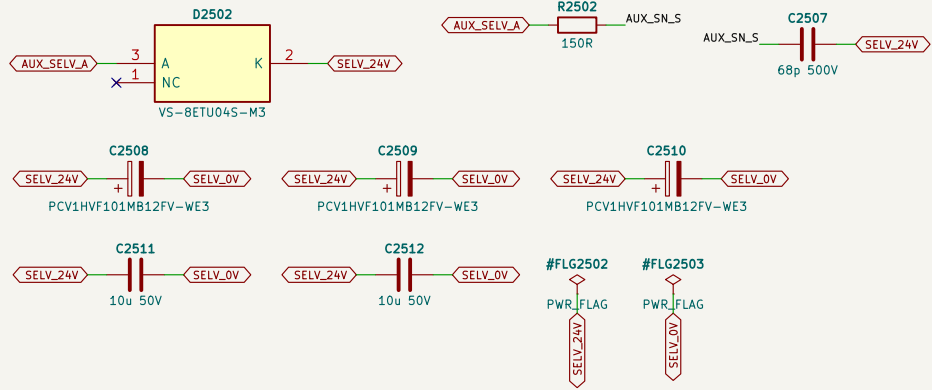
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35–80 ms -> latch -> SS crowbar -> hiccup.

LIVE 24 V RECTIFIER, HOLD-UP



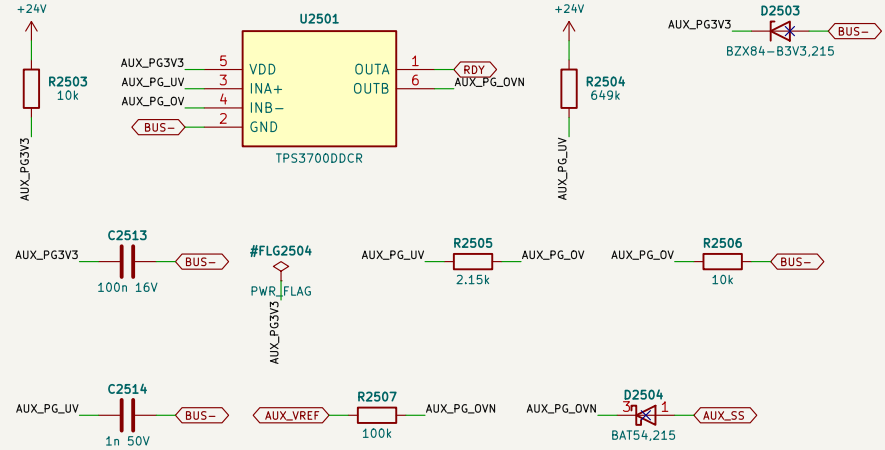
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02-21.81 V (status), OV rises 26.13-26.76 V (SS pulled low)